What Is an Exact Memory Footprint Worth to a Scheduler? Measured Footprints and Job Scheduling on Disaggregated Memory

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Abstract

A scheduler for a cluster with disaggregated memory can pack jobs only as tightly as its knowledge of each job's memory footprint allows. We measure that knowledge for 25 programs at three inputs each, two LLM inference engines among them, with the full Pin trace as the truth, MemPrint's windowed and source-only estimates, Massif, DHAT, heaptrack, the gperftools heap profiler and sampled RSS, and we replay the measured footprints through DMSched-Sim, a simulator of 32 nodes with local DRAM and a shared 1 TB pool, in which only the footprint oracle changes. Knowing a job's footprint curve rather than its peak is where most of the gain lies: at 80% offered load the exact curve gives a mean response time 26.4× shorter than the user's request and 5.3× shorter than the exact peak, and it needs 5.0 TB of memory for a service level that the exact peak meets with 6.0 TB and the request with 11.0 TB. Among measured estimators, MemPrint windowed at 5% is the only one that beats the exact peak (389 against 448 minutes), at 11 times the native run time against 227 times for the full trace. The heap profilers report the right peak for most programs but read GCC's cc1 at 0.14 to 0.16 times its peak, llama2.c, whose weights are mapped from a file, at 0.17 times and darknet at 1.97 times, so no margin makes them safe without over-reserving every other job; with the LLM engines in the corpus, the calibrated margins of heaptrack and gperftools exceed twice their estimates. On a stream of LLM jobs alone, the heap profilers fail because the weights are outside the heap, and MemPrint's curves failed because they saw the weights only in their windows; tracking the program's file mappings between windows fixes the latter (386 instead of 4,134 minutes on the LLM stream, and 327 instead of 389 on the full corpus, where every MemPrint curve then beats the exact peak). Curves built from DHAT's three recorded points do worse than DHAT's peak under every margin. When the truth is the resident set on wall-clock time, every measured curve, MemPrint's included, needs about 60% headroom and peaks do better than curves. Placing MemPrint's curve on the native clock, through hardware counts of the native run's memory references, and at the level of an earlier run's RSS cuts that headroom to 17% and its response time from 1,873 to 569 minutes, and against the resident set without code it matches the exact peak (422 against 429 minutes); a curve still gains nothing over a peak there, because the allocator keeps freed memory resident. We also evaluate three published policies, checked against their authors' code where it is published. Pond, which holds a job's request and places its predicted untouched memory in the pool, is slower than the request itself (2,519 minutes), because the pool fills first. KS+, which learns a step function over time from the program's other inputs, gives 1,262 minutes: its regressions extrapolate well for most programs but predict almost nothing for two, whose jobs then fail dozens of times. Dynamic provisioning, which resizes a running job every five minutes to the coming five minutes of its use, gives 457 minutes, between MemPrint's curve and the exact peak, but only because it reads that use from the trace, and it restarts about 300 jobs per stream; a variant that knows only the past, as a running system does, gives 504 minutes, behind both. An appendix repeats every result at offered loads from 0.4 to 1.2.

1 Introduction

A batch scheduler places a job only after it decides how much memory to reserve for it, and on a cluster with disaggregated memory that decision also sets how much of the shared pool the job takes away from other jobs. Disaggregated memory attaches a pool of memory, over CXL or RDMA, that every node can use in addition to its local DRAM [11, 23]. The pool lets a cluster provision less local memory per node, but only if the scheduler can pack jobs tightly, and it can pack them only as tightly as its knowledge of each job's memory allows. Today that knowledge is weak. Peng et al. found that over 90% of LLNL jobs use under 15% of a node's memory [15], and Zacarias et al. showed that a uniform 60% overestimate of memory demand cuts the throughput of a disaggregated system by about 25% and raises the mean response time about five times [24].

Two kinds of knowledge are possible. A *peak* is one number per job, held for the whole run. A *footprint curve* gives the memory the job needs at every point of its run, so a scheduler can let two jobs whose peaks do not overlap share the same memory. The usual sources of either are the user's request, a heap profiler such as Massif [18], DHAT [20], heaptrack [21] or the gperftools heap profiler [9], the resident set size (RSS) of an earlier run, a learned predictor [11], or a monitor that resizes a job's allocation while it runs [25]. MemPrint [14] adds two more: a footprint curve measured by instrumenting a sample of a run's memory accesses with Pin [13], and a footprint predicted from the program's source code before it runs.

However, the simulators that study scheduling on disaggregated memory either model one run in cycle detail with no scheduler [17, 4, 5, 22], or treat a job's memory as one declared peak with a synthetic error [24, 23]. None feeds the scheduler the footprints that real profilers report, and none compares those with a footprint curve known before the job starts. It is therefore unclear how much better a scheduler does with an exact footprint than with the tools that exist, and which property of the footprint (its peak, its shape over time, or how safely it bounds the truth) the gain comes from.

In this work, we measure every source of footprint knowledge listed above on the same 25 programs and replay the results through one scheduler, with everything but the footprint oracle held fixed. We make the following contributions:

- We build a corpus of footprint curves for 25 programs at three inputs each, HPC kernels, interpreters, a compiler and two LLM inference engines among them, from the full Pin trace (the truth), MemPrint's windowed and static estimates, Massif, DHAT, heaptrack, gperftools and sampled RSS, together with what each method costs (Section 4).

- We build DMSched-Sim, a cluster simulator with local memory and a shared pool, an EASY-backfilling scheduler that admits a job only if its whole reservation curve fits, a remote-memory slowdown model and OOM kills (Section 3).

- We show how much of the gain an exact footprint brings comes from knowing the curve rather than the peak, how close each measured estimator gets, how much memory each needs for the same service, and what each estimator's cost buys (Section 5).

- We evaluate curves derived from DHAT, which records no time series, and show why its three recorded points do not substitute for a measured curve (Section 6).

- We schedule a stream of LLM inference jobs alone, whose memory is mostly model weights mapped from a file, show which estimators survive it (Section 5.6), and extend MemPrint's windowed mode to track file mappings between windows, which removes its failure there (Section 5.8).

2 Background and related work

Disaggregated memory simulators. DRackSim [17], CXL-ClusterSim [4], CXL-DMSim [5] and CXLMemSim [22] model the timing of accesses to pooled memory for one run, in cycle detail or from a trace, and none of them has a job scheduler. The BSC Slurm simulator extended by Zacarias et al. schedules real job traces on a disaggregated system [23], but it represents a job's memory as one peak, and its study of accuracy multiplies that peak by a synthetic overestimate [24]. Their dynamic provisioning study resizes each job's allocation every five minutes to its use, read from usage traces, and fails and restarts a job whose growth does not fit [25]. That is a monitor that acts while the job runs, not knowledge available before it starts, and we evaluate it as a baseline. Boezennec et al. study the allocation of disaggregated memory to jobs with piecewise-constant memory phases [3], CFM schedules jobs against per-job slowdown curves measured offline [1], and Pond predicts the untouched memory of virtual machines with a gradient-boosted model, places it in a CXL pool and corrects overpredictions with a QoS monitor [11]. We evaluate it as a baseline as well. KS+ predicts the memory of workflow tasks over time, as a step function whose segments it learns from earlier executions with other inputs, and reduces wasted GB-seconds by 51% against peak-only prediction [2], but it has no cluster scheduler or memory pool. We evaluate it as a third published baseline.

Footprint estimators. Massif samples the heap at up to 100 snapshots by default, on a time axis of instructions or of milliseconds under Valgrind [18], and with `-pages-as-heap` it counts mapped pages instead of heap blocks. DHAT reports the heap at its global peak (t-gmax) and at exit, and the time of the peak, but no time series [20]. heaptrack records every allocation and can export a heap curve [21], and the gperftools heap profiler writes a dump each time the heap grows by a set amount [9]. All four count requested heap bytes, so they miss memory a program maps itself and count blocks that are allocated but never touched. MemPrint counts the bytes a program actually touches, from a sample of its memory accesses [14]. Its windowed mode watches accesses only in short windows (1% or 5% of the run) and tracks allocations and page residency in between, and its static mode predicts the footprint from the source code with an interpreter that counts the references a build of the program makes.

3 Simulator

DMSched-Sim replays one stream of jobs once per footprint oracle, so that any difference between two runs comes from what the scheduler knows about memory. It is a time-slotted simulator (one slot is one minute) written in Python (`analysis/memprint/sched/`).

System model. The cluster has 32 nodes, each with 32 cores and 128 GB of local DRAM, and one shared pool of 1 TB (one sixth of all memory). A job runs on one node. Its memory is placed in the node's free local memory first, and the rest goes to the pool.

Jobs. A job has an arrival time, a core count, a base runtime (its runtime with all memory local), a true footprint curve $a(p)$ over its progress $p \in [0, 1]$, and a reservation curve $r(p)$ set by the oracle. Curves are defined over progress rather than time, so a job slowed by remote memory stretches its curve in wall-clock time.

Scheduler. The scheduler is first come, first served with EASY backfilling [12] over a window of 32 queued jobs. Each node keeps a timeline of committed local memory and cores, and the pool keeps one of committed memory. A job starts on a node only if, at every slot from now to its projected end, the part of its reservation that does not fit in the node's free local memory fits in the free pool. With a time-varying reservation, two jobs whose peaks do not overlap can therefore share the same memory. When the head of the queue cannot start, the scheduler reserves its earliest feasible start, and a later job may start early only if it fits beside that reservation. Among the nodes that fit, the scheduler picks the one with the least pool use and then the least free local memory left over (best fit).

Remote-memory slowdown. Following the linear models of CFM [1] and Boezennec et al. [3], a job's runtime is stretched by the share of its resident GB-minutes that lie above its local reservation:

$$T = T_{\text{base}}\,(1 + \beta f_{\text{remote}}), \qquad f_{\text{remote}} = \frac{\sum_t \max\bigl(0,\, a(t) - \ell(t)\bigr)}{\sum_t a(t)}, \tag{1}$$

where $\ell(t)$ is the local part of the reservation and $\beta = 0.5$, so a job whose memory is entirely remote runs 1.5 times as long. Because the runtime depends on the placement and the placement depends on the runtime, the two are solved together by fixed-point iteration when the job is admitted. The scheduler ranks candidate nodes on the job's base runtime and then solves the stretch on the best node, falling back to the next node if the stretched job no longer fits.

OOM kills. If the true footprint exceeds the reservation at any slot, the job is killed there, its work is lost, and it is requeued at the head of the queue with its reservation multiplied by 1.5, as a batch system that raises a job's memory limit after an out-of-memory failure would do. A reactive oracle (Section 3.3) is the exception: its growth past the reservation is granted from free local memory or the pool while either has room, and the job is killed only when neither does.

Metrics. We report the mean response time (completion minus arrival), the mean bounded slowdown (response over $\max(T_{\text{base}}, 10\,\text{min})$), the number of OOM kills, and the reserved and the actually resident memory as shares of all memory. To leave out the warm-up and the drain of a finite stream, every metric is taken over the jobs that arrive between the 10th and the 90th percentile of arrival times, and the memory shares are averaged over that period.

Validation of the simulator. Fifteen unit tests (`tests/sched/test_sim.py`) check that two jobs with non-overlapping peaks share a node under their curves but run one after the other under their peaks, that an under-reservation is killed where the footprint passes it and requeued, that backfilling never delays the head of the queue, that a fully remote job runs $1+\beta$ times as long, that elastic growth is granted only from free memory, that Pond's untouched part goes to the pool and its QoS monitor shortens an overpredicted job's stretch, and that a dynamic resize that no longer fits restarts the job. On the synthetic stream of the earlier prototype (3,000 jobs, five footprint shapes, a 60 GB median heap) the simulator reproduces the prototype's ordering at saturation: the exact curve sustains 101 jobs per hour, the exact peak 16% fewer, a Massif-style curve 37% fewer and a DHAT-style peak 57% fewer.

3.1 Job streams

The main experiments use a controlled stream of 3,000 jobs in which memory is the contended resource. Each job has 1 to 16 cores, a lognormal base runtime with a median of 30 minutes

clipped to 5 to 720 minutes, and a lognormal memory request with a median of 100 GB. What a job really uses is its request multiplied by a ratio $u \leq 1$ taken from the LANL CM5 log of the Parallel Workloads Archive [6], the archive log that records both requested and used memory (median $u = 0.79$ over its 94,335 jobs with both fields). The shape of the job's footprint, and what every estimator reports for it, come from one corpus program at its largest input (Section 4), chosen at random. All of that program's curves are scaled by the factor that puts the truth's peak at the job's peak, so each estimator keeps its own over- or under-estimate. Arrivals are spread evenly in random order over a span set so that the *offered load*, the resident GB-minutes arriving per minute over the cluster's total memory, takes the values 0.4, 0.6, 0.8, 1.0 and 1.2. Each point is three seeds.

We also tried the MetaCentrum2 log [6] (2013 to 2015, 5.7 million jobs with a memory request but no used memory). In a window of 3,000 single-node jobs, 1% of the jobs carry 92% of the requested GB-minutes, so the results of a replay depend on a handful of very large jobs. We therefore use the controlled stream for the main results.

A real trace with measured memory. The SURF Lisa trace [19] joins 1.6 million Slurm jobs of a 343-node cluster with their nodes' Prometheus metrics, sampled every 30 s from June to November 2022. A node's memory is a job's only while the job has the node to itself, so we keep a job only if no other job shared any of its nodes at any sample and at least 80% of the samples its runtime implies are present: 88,767 jobs of at least five minutes, 1,429 of them on 2 to 22 nodes. A job's memory is its nodes' active memory above each node's idle level (its 1st percentile), resampled to the job's progress, the largest over its nodes at each point; most jobs take whole 16-core nodes and peak at a few GB. We replay windows of 3,000 consecutive jobs in two ways. In the *measured* stream every job keeps its own curve, and only the oracles that need nothing but the job's own memory apply. In the *hybrid* stream every job keeps the trace's arrival, runtime, node count and measured peak, with a corpus program's footprint shape scaled to that peak, so that every estimator applies. Lisa records no memory request; a job requests Slurm's default on the simulated cluster, the node's memory per core times the cores it uses, and at least its peak. Both streams run on a Lisa-shaped cluster of 32 nodes with 16 cores each, a pool of a quarter of the local memory, and local memory per node swept from Lisa's 103 GB down to 4 GB; the offered load is set against each cluster's own memory, so a policy's results are compared with the other policies' at the same memory, not across memories. A job on several nodes holds its footprint on each, its nodes share the pool, and it runs as long as its slowest node.

3.2 Two truths

MemPrint counts the bytes a program touches, but a kernel charges and OOM-kills a process for its resident pages. The two differ by page granularity, by memory an allocator keeps after a free, and by the runtime's own pages (loader, libraries, MPI, an interpreter). We therefore run every experiment with two truths. With *bytes touched* as the truth, the scheduler must cover the footprint a page-perfect system would need. With *RSS* as the truth, it must cover the resident set sampled every 5 ms. The runtime's own pages do not grow with a job's data, so under the RSS truth only the RSS above the run's smallest sample (1.2 to 9.5 MB across the corpus) is scaled to the job's size.

Table 1: Footprint oracles. "Curve" oracles follow a measured footprint over the job's progress; "peak" oracles hold one value for the whole run.

Oracle	Kind	Source
Exact curve	curve	the truth, at full resolution (the upper bound)
Exact, 20 windows	curve	the truth, coarsened as the measured curves are
Exact peak	peak	the truth's peak
MemPrint windowed 1%, 5%	curve	MemPrint watching 1% or 5% of the run
MemPrint pages only	curve	MemPrint page residency, nothing watched
MemPrint static	curve	predicted from source before any run
Massif curve, pages-as-heap	curve	Massif snapshots (heap, or mapped pages), instruction clock
Massif, ms clock	curve	the same, with time in milliseconds under Valgrind
heaptrack curve	curve	heaptrack's heap over time
DHAT peak	peak	heap at DHAT's global peak (t-gmax)
DHAT curve, hold-then-decay	curve	three points recorded by DHAT (Section 6)
gperftools peak	peak	largest in-use heap of the heap profiler's dumps
RSS peak	peak	peak RSS of an earlier run of the same input
User request	peak	the job's memory request
ML peak	peak	request $\times$ a 0.95-quantile gradient-boosted prediction of u
Reactive resizing	curve	request for 5 minutes, then the last 5 minutes' largest use + headroom
Pond	peak	the request, its predicted untouched part in the pool; QoS monitor
Dynamic provisioning	curve	request, then each 5 minutes the coming 5 minutes' largest use
Dynamic, look-back	curve	request, then the past 5 minutes' largest use, grown on demand
KS+	curve	4 segments regressed on input size from the other inputs' RSS; retry

3.3 Oracles and margins

An oracle turns what is known about a job into its reservation curve. Table 1 lists them. Curve oracles follow a measured curve, coarsened to 20 windows over progress, each holding its largest value and widened to its neighbours, because each method's clock (memory references for MemPrint, instructions or milliseconds under Valgrind for Massif, seconds for RSS) maps to wall-clock progress only approximately. Peak oracles hold one value for the whole run.

The ML oracle follows Pond [11]. It predicts u from the request, the core count, the requested time and the user's and the executable's median u in their history, with a gradient-boosted model at the 0.95 quantile, trained on half of the CM5 users. The jobs it predicts take their u from the other half of the users. The reactive oracle is a simplified form of the dynamic provisioning of Zacarias et al. [25] that sees only the past. The scheduler is shown its future resizes when it admits the job, which favours this baseline.

Two further baselines follow the two published systems as closely as the simulator allows. The *Pond* oracle [11] reserves the job's whole request, so it never causes an OOM kill, but it keeps only the request minus the predicted untouched memory in local DRAM. The rest, rounded down to whole GB, is placed in the pool, as Pond's zero-core NUMA node does for a virtual machine, even when local memory is free. The untouched share is predicted by the same gradient-boosted model as the ML oracle, at the 0.98 quantile, which corresponds to Pond's target of 98% of jobs within their performance margin. A job that touches more than it was given locally runs on the pool. Five minutes after it first does so, Pond's QoS monitor moves its pool part to local memory, where the node has room. Pond's second model, which places latency-insensitive virtual machines entirely in the pool, needs hardware performance counters that the job streams do not have, and is left out. The *dynamic provisioning* oracle follows the Slurm policy of Zacarias et al. [25]. A job starts with its request. Every five minutes, its allocation is resized to the largest use of the coming five minutes,

which their simulator reads from the usage trace. The resize takes local memory first and the pool next, with no headroom. Unlike the reactive oracle, a resize is committed only when it happens, so another job may take the memory first. A resize that does not fit then fails the job, which restarts from the beginning (their fail/restart policy), and we count it with the OOM kills. A running system cannot read the coming interval, so we also evaluate a *look-back* variant: each update sets the allocation to the previous five minutes' largest use, the job grows on demand whenever its use exceeds the allocation, from free local memory and then the pool, and memory is committed a minute at a time. A growth that does not fit fails the job as before; after three restarts it runs with its request held, as Zacarias et al. suggest for jobs that fail repeatedly, because otherwise a job whose growth the scheduler keeps giving to newer jobs restarts indefinitely. Its runtime stretch is fixed when it is admitted, from the placement it would get then. Neither oracle uses a margin. We checked both against the authors' artifacts. Pond's artifact contains only its CXL emulation scripts, not its prediction models, QoS monitor or scheduler, so our Pond oracle follows the paper's description. The dynamic-provisioning code in the authors' Slurm simulator (branch `dynamic_mem_reschedule_ov_bj`) matches our oracle in admission against current allocations only, remote-first release and local-first growth, and restart from the beginning with the original request; it differs in recomputing the remote-memory slowdown at every resize and in charging 10 s of progress for every resize that grows, both of which we leave out, in favour of the baseline. Which trace value a resize applies is set by a preprocessing step that is not published; we follow the paper.

The *KS+* oracle follows Bader et al. [2] and the authors' code. KS+ learns a task's memory over time from earlier executions of the same task with other inputs. It cuts each execution's monitored memory into four segments of a non-decreasing step function, and fits, for each segment, a linear regression of its end time and one of its peak on the input size. For a new input it ends every segment 15% earlier than predicted and raises every peak by 10%, with peaks of at least 100 MB. After an OOM kill, it ends the failed segment, and proportionally the later ones, at the time of the failure, or raises the last segment's peak by 20% if the failure was there, and runs the job again. We give KS+ the RSS of each program's native runs at its other corpus inputs, on their wall clocks, and the size of the data each input describes (array elements, vertices, grid points, the source file, the model file). A job, at the program's largest input, is predicted from the two smaller ones; on the LLM stream, every input is predicted from the other two of its engine. darknet has a single input and no history, so KS+ reserves its request. KS+ uses its own offsets and no further margin.

Every estimate needs a margin, because an estimate below the truth causes OOM kills. A curve oracle's margin is headroom added as a share of its own peak, since a curve on another clock can be low where the truth is not, and no factor lifts a zero. A peak oracle's margin multiplies its peak. We use four margin policies: a fixed 10% and 30%, and two *calibrated* margins that a site could set from its own history. For each oracle and program we compute the smallest margin with which the oracle never falls below the truth. A job then receives the 90th percentile (*calibrated 90%*) or the largest value (*calibrated 100%*) of the other programs' margins, so no oracle is tuned on the program it predicts. The request, ML and reactive oracles have no profile to calibrate, and they keep a 10% margin.

4 Footprint corpus

We ran 25 programs at three inputs each, from the smallest to the largest, on a 32-core node with 62 GB of memory (Table 2): four PolyBench kernels [16] (2mm, gemm, jacobi-2d, atax, unoptimised builds at SMALL, MEDIUM and LARGE), miniVite [8] with four threads, the GAP benchmarks bfs, pr, cc and sssp on Kronecker graphs of 2^{16} to 2^{18} vertices, Lua, Python, Perl and SQLite

Table 2: The corpus at each program's largest input: the true peak (bytes touched), the mean over the run as a share of the peak, each method's peak as a share of the true peak, and the time each method takes over the native run.

		Footprint (ratios to the true peak)								Cost (time / native)			
Program	Input	Peak (MB)	Mean/peak	RSS	MemPrint W5	Massif	DHAT	heaptrack	Native (s)	W5	Massif	DHAT	Full trace
2mm	LARGE	37.1	0.96	1.03	1.00	1.00	1.00	1.00	8.6	4.8	5.6	15.2	357
atax	LARGE	32.1	0.87	1.04	1.00	1.00	1.00	1.00	0.1	21.8	5.7	9.8	122
GCC cc1	large	105.5	0.69	1.27	1.06	0.16	0.15	0.15	1.8	100.5	11.0	68.1	206
darknet	alexnet	269.9	0.86	1.07	1.04	1.97	1.97	1.97	3.0	11.8	10.5	45.9	773
GAP bfs	18	72.8	0.36	1.05	1.00	0.99	0.99	0.99	1.5	8.1	3.0	5.7	83
GAP cc	18	72.0	0.36	1.06	1.01	1.01	1.01	1.01	2.2	5.5	2.2	3.9	56
GAP pr	18	72.4	0.40	1.05	1.01	1.00	1.00	1.00	2.8	5.2	2.3	5.3	123
GAP sssp	18	135.8	0.36	0.95	1.00	1.00	1.00	1.00	2.5	5.9	2.4	4.4	116
gemm	LARGE	29.1	1.00	1.04	1.00	1.00	1.00	1.00	6.1	4.4	5.7	15.2	356
HPCCG	64	102.7	0.99	1.03	1.00	1.02	1.02	1.02	1.7	13.3	10.5	24.8	310
jacobi-2d	LARGE	27.1	1.00	1.04	1.00	1.00	1.00	1.00	12.8	3.7	7.3	16.2	275
Kripke	16	274.6	0.88	1.02	1.00	1.00	1.00	1.00	3.4	11.4	4.8	50.7	473
llama2.c	110M	449.7	0.98	1.14	1.00	0.17	0.17	0.17	14.9	3.5	3.6	20.5	41
llama.cpp	135m-n128	352.9	0.89	1.08	1.00	1.05	1.05	1.05	1.5	47.1	137.1	279.6	604
Lua	1	57.1	0.53	1.15	0.94	1.02	0.92	0.92	4.2	14.8	4.3	28.0	161
LULESH	60	151.1	0.62	1.02	1.00	1.00	1.00	1.00	5.5	10.6	7.9	32.2	458
miniVite	32768	15.4	0.51	1.99	1.00	1.55	1.53	1.42	1.8	42.5	22.9	36.9	322
MiniFE	80	256.5	0.98	1.04	1.00	1.03	1.02	1.02	9.2	10.7	6.7	23.8	273
NPB CG	A	27.4	1.00	1.10	1.00	0.99	0.99	0.99	1.9	10.6	7.8	15.5	237
NPB IS	A	69.7	0.81	1.03	1.00	0.99	0.99	0.99	2.4	3.9	2.7	7.0	61
NPB MG	W	58.5	0.98	1.04	1.00	0.99	0.99	0.99	0.5	15.5	4.0	8.8	107
Perl	1	56.1	0.67	1.15	0.97	1.04	0.97	0.96	1.8	30.9	5.2	44.8	145
Python	1	95.9	0.37	1.19	1.04	1.29	1.01	0.14	5.6	17.0	12.5	175.5	272
Quicksilver	100000	42.2	1.00	1.28	1.01	1.02	1.01	1.01	5.5	13.9	6.3	44.5	227
SQLite	1	77.8	0.43	1.06	0.97	1.01	1.01	1.01	1.2	47.7	7.1	39.0	204

workloads at three sizes, GCC's cc1 compiling three source files at `-O0`, HPCCG, LULESH, NPB IS, CG and MG (classes S, W and A), MiniFE, Quicksilver, Kripke, darknet classifying images with AlexNet (one input), and two LLM inference engines: llama2.c [10] generating 128 tokens from "Once upon a time" with the stories15M, 42M and 110M models, and llama.cpp [7] (b11500, CPU, four threads) completing the same prompt with SmolLM2-135M (32 and 128 tokens) and SmolLM2-360M (64 tokens), quantised to 8 bits, at the model's default 8192-token context. Both engines map their weights from the model file rather than allocating them, and llama.cpp sizes its KV cache for the whole context and clears it with `memset`. The native runs take 0.12 to 14.9 s, and the true peaks at the largest inputs are 15 to 450 MB.

Every (program, input) ran under ten methods: the program alone, with its RSS sampled every 5 ms; the full Pin trace (MemPrint's splitter with frees tracked), which gives the bytes touched over time and is the truth; MemPrint windowed at 1% and 5% watched and pages only, with the tool built from the `windowed-next` branch; Massif and Massif with `-pages-as-heap` (Valgrind 3.22, up to 1,000 snapshots), each run twice, with Massif's default clock of instructions executed and with milliseconds under Valgrind; DHAT; heaptrack 1.3.0; and the gperftools 2.18 heap profiler with a dump every 1/50 of the native peak RSS. The MemPrint static predictions come from the C interpreter for PolyBench (a curve over time) and from the C++ interpreter for HPCCG, LULESH and NPB (a whole-run footprint). The NPB predictions are those of the second blind test, made before any trace of NPB existed; the static route does not cover the LLM engines. Of the 874 runs, two failed: the full traces of NPB MG class A and of llama.cpp with the 360M model ran out of memory (the latter at 20.7 GB, in Pin's allocator), so MG is used up to class W and llama.cpp up to the 135M model with 128 tokens. The runs shared the node with up to 15 others (the LLM runs with up to 7), so the costs below are slowdowns under load.

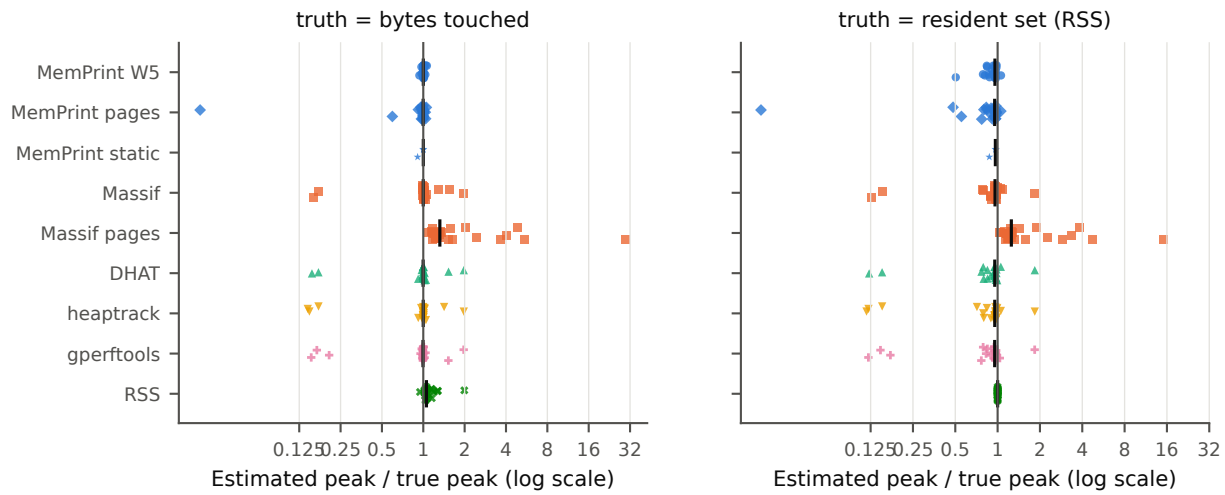


Figure 1: Each method's peak over the true peak, every program at its largest input (one marker per program, the black bar is the median). Left: against bytes touched. Right: against the RSS.

4.1 Accuracy of the peak

Most heap profilers report the right peak for most of these programs, and fail badly on the rest (Figure 1). Specifically, Massif's peak is within 5% of the true peak on 20 of the 25 programs, DHAT and gperftools on 20 and heaptrack on 19, because these programs touch nearly all the heap they allocate. However, the heap profilers read 0.14 to 0.16 times the peak of GCC's cc1, whose garbage collector maps its own pages, and 0.17 times that of llama2.c, whose 110M model's weights are mapped from the file and never pass through the heap; they read 1.97 times the peak of darknet and 1.42 to 1.55 times that of miniVite, which allocate memory they never touch. On llama.cpp they are within 5% (1.04 to 1.05), because its weights' file mapping is offset by a KV cache that it allocates on the heap and clears in full. MemPrint windowed at 5% is within 5% on 23 of the 25 programs (0.94 to 1.06 times the truth). Pages only is within 5% on 21; it estimates the resident pages of heap blocks and writable sections, so it misses the read-only weight mappings and reads llama2.c at 0.02 and llama.cpp at 0.60. Massif with `-pages-as-heap` overestimates every program (1.07 to 29.9 times), and the RSS is 0.95 to 1.99 times the bytes touched (median 1.05).

4.2 Shape over time

The footprint of half of the programs varies over the run. The mean over the run is below 70% of the peak for 11 of the 25 programs (GCC's cc1, the four GAP kernels, Lua, LULESH, miniVite, Perl, Python and SQLite), and it is 0.36 to 0.40 times the peak for the four GAP kernels. A scheduler that holds the peak for such a job reserves memory that the job leaves unused for most of its run. Figure 2 shows Python's footprint, which rises and falls seven times. MemPrint windowed at 5% follows it within a few percent. Massif's curve, on its clock of instructions, follows its phases but sits above it, because Massif counts requested heap bytes and Python allocates memory it never touches (1.29 times the true peak). DHAT records only the peak, its time and the heap at exit, so a curve built from those points misses every phase but one.

The LLM engines differ from the rest of the corpus in where their memory is (Figure 3). Both map the model's weights from the file, and both read all of them for the first token, so the true footprint reaches 90% of its peak within the first 1 to 2% of a llama2.c run and stays there (mean

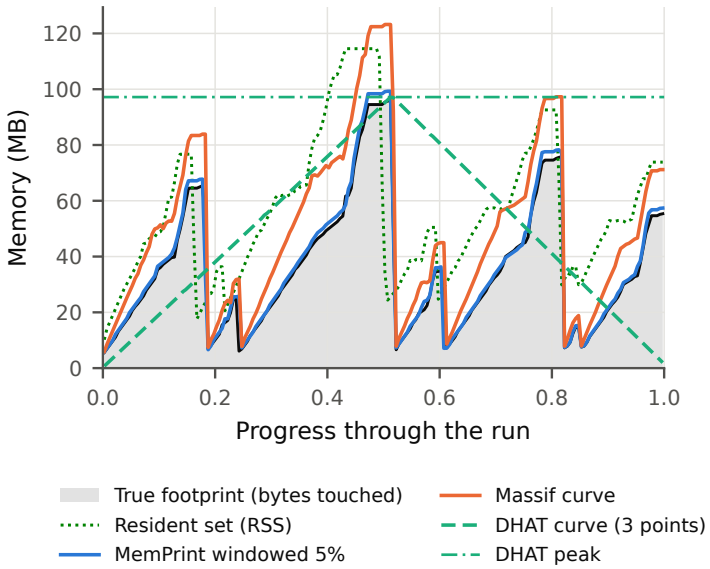


Figure 2: Python (largest input): the true footprint over the run, the RSS, and the curves and peaks the estimators report. Each curve is on its own method's clock, rescaled to progress.

0.96 to 0.98 of the peak). llama.cpp reaches it at 12% of a 128-token run, and its mean is 0.89 of the peak. The heap profilers see only the heap: 0.08 to 0.21 of llama2.c's peak, and the KV cache and compute buffers of llama.cpp. MemPrint windowed at 5% reaches about the right peak (0.86 to 1.00 over the five configurations), but late: its curve reaches 90% of the peak at 19 to 20% of a llama2.c run and at 56% of the llama.cpp run, because between its windows it tracks heap blocks and writable pages, not a read-only file mapping, so it learns of the weights only when a window watches the program reading them. Pages only, which never watches, reads llama2.c at 0.02 to 0.05 of its peak.

4.3 Cost

MemPrint's windowed estimates are not cheaper than the heap profilers on these short runs (Figure 4). At the largest inputs, the median slowdown is 1.4× for heaptrack, 1.2× for gperftools, 5.7× for Massif (6.4× with its clock in milliseconds) and 23.8× for DHAT, against 6.3× for MemPrint pages only, 8.9× at 1% watched and 11.4× at 5% watched. The full Pin trace costs 226.8× (41.2 to 772.5×). The two LLM engines sit at the ends of that range. llama2.c, a single-threaded run of 2 to 15 s, costs 41 to 49× under the full trace, 3.5 to 6.6× under MemPrint at 5% and 19 to 21× under DHAT. llama.cpp, a sub-second run with four threads, costs 469 to 604× under the full trace, 47 to 91× under MemPrint at 5%, 82 to 137× under Massif and 182 to 280× under DHAT, while heaptrack and gperftools stay at 2.4 to 5.8×. Pin's fixed start-up cost weighs heavily on runs of a few seconds, and the windowed mode is slowest on programs with much code to re-translate (GCC's cc1, 93.6× at 1%). The static route needs no run of the program, but its interpreter takes 7 to 59,000 times the native run. It predicts atax MEDIUM in 0.8 s, the other PolyBench kernels at MEDIUM in 13 to 24 s and HPCCG in 255 to 989 s, but NPB IS and CG at class W take 4.0 and 5.5 hours.

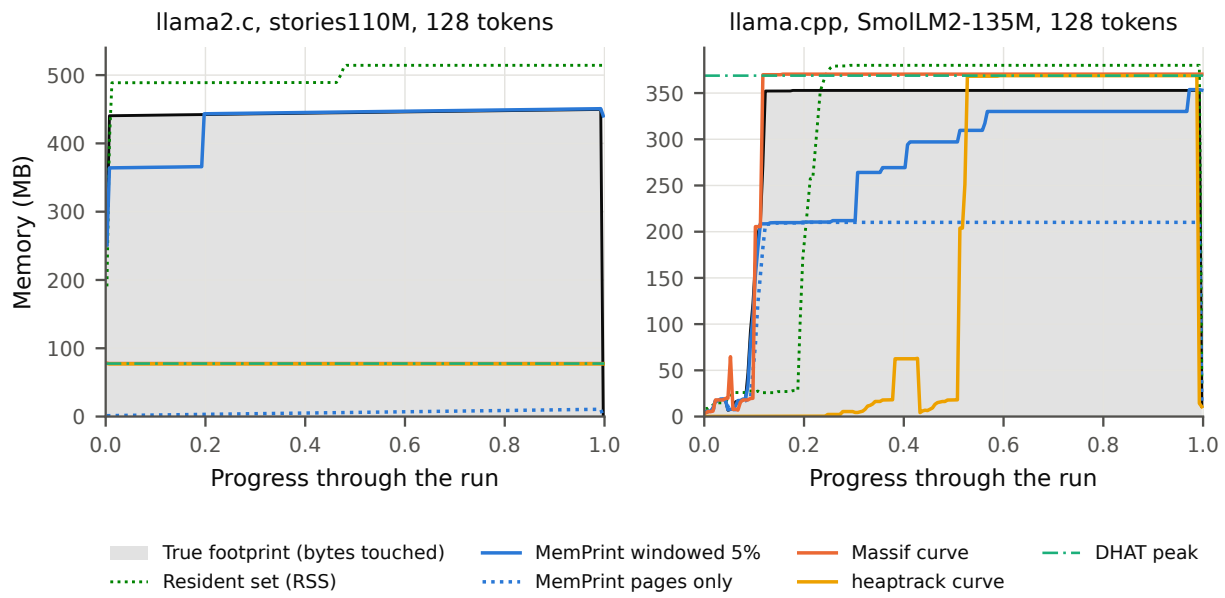


Figure 3: The two LLM engines at their largest inputs: the true footprint over the run, the RSS, and the curves and peaks the estimators report, each on its own clock rescaled to progress.

4.4 Static predictions

The static predictions are within 0.46% of the true peak at the median, over the 19 (program, input) pairs they cover. They fail in two known ways. For LULESH, the static route predicts the footprint of the whole run, 1.57 times the live peak, because LULESH frees and reallocates its arrays between iterations and the static route does not model frees for C++ programs. For NPB MG, the prediction is 7 to 9% short, as it was in the blind test, for a reason we have not found.

5 Results

We first ask how much an exact footprint is worth and where the gain comes from, then how close each measured estimator gets, and finally what each estimator's cost buys. Table 3 gives every oracle at offered load 0.8 with calibrated 90% margins, and Figure 5 gives the response time across loads. Appendix B repeats every result of this section at offered loads 0.4, 0.6, 1.0 and 1.2.

5.1 An exact footprint curve

Knowing a job's footprint curve, rather than its peak, is where most of the gain lies. At load 0.8 the exact curve gives a mean response time of 84 minutes, against 448 minutes for the exact peak and 2,208 minutes for the user's request, under the bytes-touched truth, which makes the curve 26.4× better than the request and 5.3× better than the peak, and the peak 4.9× better than the request. The RSS truth gives the same picture (99, 423 and 2,183 minutes). The curve lets the scheduler keep 81% of all memory resident with jobs, against 58% for the exact peak and 27% for the request, because jobs whose peaks fall at different times share the memory that a peak would hold idle. Coarsening the exact curve to the 20 widened windows used for every measured curve raises the response to 173 minutes, so about three quarters of the curve's advantage over the peak survives the resolution and the timing slack a measured curve needs. The gap widens with load

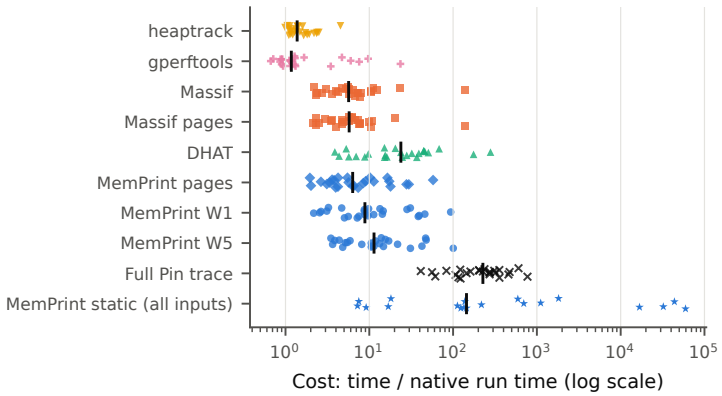


Figure 4: Each method's time over the native run, every program at its largest input (static: every input with a prediction). The black bar is the median.

(Figure 5). At load 0.4 every oracle that does not over-reserve keeps up with arrivals, and from load 0.6 the peak and the request saturate memory while the curve does not.

5.2 Measured estimators

MemPrint windowed at 5% is the only measured estimate that beats the exact peak, and only under the bytes-touched truth. With calibrated 90% margins it gives 389 minutes, against 448 minutes for the exact peak, 469 for MemPrint pages only, 580 for reactive resizing, 765 for MemPrint windowed at 1%, and 985 and 1,011 minutes for the RSS and DHAT peaks. The LLM engines cost the other two MemPrint curves their place: without them, pages only and windowed at 1% gave 381 and 358 minutes, but their curves lag the weights the engines read at the start (Section 4), and the margin that covers 90% of the programs rises from 4 to 8% for pages only and from 5 to 14% for windowed at 1% (Table 4), with 1,117 to 1,901 OOM kills per stream. The MemPrint 5% curve still loses to the exact curve by a factor of 4.6, for two reasons. First, the 20-window coarsening alone costs a factor of 2.1 (from 84 to 173 minutes). Second, a calibrated 90% margin leaves the programs that need more than the 90th-percentile margin under-reserved, which causes 461 OOM kills per stream. With a fixed 10% margin it gives 470 minutes and 216 kills.

Massif reports the heap over the whole run, not only its peak, and its curve follows a program's phases on its default clock of instructions executed. With calibrated 90% margins it gives 2,399 minutes, worse than the user's request, because the headroom that covers 90% of the programs is 52% of its peak (Table 4). Three kinds of program set that headroom. Massif reads GCC's cc1 at 0.16 times its peak, because the compiler's garbage collector maps its own pages, and llama2.c at 0.17, because its weights are a file mapping, and on the four GAP kernels its clock of instructions drifts from the truth's clock of memory references enough to need 12 to 53% headroom. With a fixed 10% margin, which leaves those jobs to be killed (1,714 kills per stream), Massif gives 991 minutes, close to DHAT. The clock matters. With Massif's clock in milliseconds under Valgrind, which includes Valgrind's start-up and runs at a different pace from the native program, the headroom rises to 100% and the response time to 4,268 minutes. On atax, for example, the heap reaches a tenth of its peak at 48% of Massif's millisecond clock but at the start of the native run. Massif with `-pages-as-heap` counts mapped pages, 1.32 times the true peak at the median, and gives 4,670 minutes. The heap profilers' peaks (DHAT, gperftools) are within 5% of the true peak for 20 of the 25 programs, so they come close to the exact peak for most jobs, but they read GCC's cc1 at 0.14 to

Table 3: Every oracle at offered load 0.8, calibrated 90% margins (10% for the request, ML and reactive oracles, which have no profile to calibrate; none for Pond and dynamic provisioning, whose Kills column counts failed resizes), mean of three seeds. Resp. is the mean response time, BSLD the mean bounded slowdown, Kills the OOM kills per stream of 3,000 jobs, Res. and Used the memory reserved and actually resident as a share of all memory.

	Truth = bytes touched					Truth = RSS				
Oracle	Resp. (min)	BSLD	Kills	Res. %	Used %	Resp. (min)	BSLD	Kills	Res. %	Used %
Exact curve	84	2.4	0	81	81	99	3.0	0	80	80
Exact, 20 windows	173	6.3	0	80	72	160	5.7	0	80	73
MemPrint windowed 5%	389	15.7	461	77	58	1869	80.5	309	58	33
Exact peak	448	18.3	0	78	58	423	17.2	0	77	61
Dynamic provisioning (5 min)	457	18.9	307	74	63	395	16.1	265	75	64
MemPrint pages only	469	19.2	1901	74	54	2516	109.0	1494	52	26
Dynamic provisioning, look-back	504	21.1	444	73	61	493	20.5	478	72	61
Reactive resizing (5 min)	580	24.2	19	71	54	612	25.6	21	70	54
MemPrint windowed 1%	765	32.2	1117	66	43	2340	101.3	620	54	28
MemPrint windowed 5%, peak	818	34.4	379	72	49	959	40.6	492	71	47
RSS peak (earlier run)	985	41.8	474	69	44	423	17.2	0	77	61
DHAT peak	1011	42.9	1323	70	45	1302	55.6	1310	65	41
KS+ (segments from other inputs)	1262	54.3	8189	59	38	844	35.7	8534	67	48
ML peak (quantile GBM)	1991	86.0	115	61	29	1992	86.1	115	61	31
User request	2208	95.6	0	57	27	2183	94.4	0	57	29
Massif curve	2399	103.8	1107	53	26	2099	90.6	1184	56	30
Pond (untouched in pool)	2519	109.0	0	53	26	2472	106.9	0	53	27
DHAT curve (3 points)	3512	152.9	1228	51	23	3008	130.7	1157	52	25
DHAT hold-then-decay	4161	180.9	1144	48	20	3943	171.4	1152	47	21
Massif curve, ms clock	4268	185.9	1056	45	17	2684	116.3	1190	51	26
Massif pages-as-heap	4670	203.7	668	40	15	4472	194.4	692	42	16
Massif pages-as-heap, ms clock	6619	290.0	465	35	10	5669	247.5	697	37	12
gperftools peak	10476	458.4	1644	33	7	11181	490.8	1644	33	7
heaptrack curve	11778	516.0	1285	31	6	12610	552.6	1439	31	6

0.16 times its true peak and llama2.c at 0.17, and those jobs are killed and restarted repeatedly. For DHAT, these two programs fall above the 90th percentile of the margins, so its calibrated margin stays at 6% and it gives 1,011 minutes with 1,323 kills per stream. For gperftools and heaptrack a third program falls there as well, Python, whose heap they read at 0.21 and 0.14 times its peak, so the margin that covers 90% of the programs becomes 233% and 326% of the estimate, every job reserves three to four times its estimate, and the response time rises to 10,476 and 11,778 minutes, about five times the request's. With a fixed 10% margin, which accepts those kills, they give 1,158 and 1,154 minutes. The ML peak (1,991 minutes) improves little on the request it scales, because its 0.95-quantile prediction of u covers only 76% of jobs from unseen users. Reactive resizing (580 minutes) is the best oracle that needs no profile, although it is shown its future resizes when the job is admitted.

Pond and dynamic provisioning. Dynamic provisioning gives 457 minutes under the bytes-touched truth, between MemPrint windowed at 5% (389) and the exact peak (448), and 395 minutes under the RSS truth, where it beats every oracle but the exact curve, the exact peak included (423). It pays with 307 and 265 failed resizes per stream, each a restart from the beginning. Two things make it this good, and both come from its simulation rather than from what a running system could know. Each resize is the largest use of the coming five minutes, read from the truth, so between resizes it is an exact curve at a five-minute resolution, with no margin and on the wall clock. And a job's first five minutes, at its full request, are a small share of runs with a median of 30 minutes. The look-back variant, which knows only the past as a running system does, gives 504 and 493

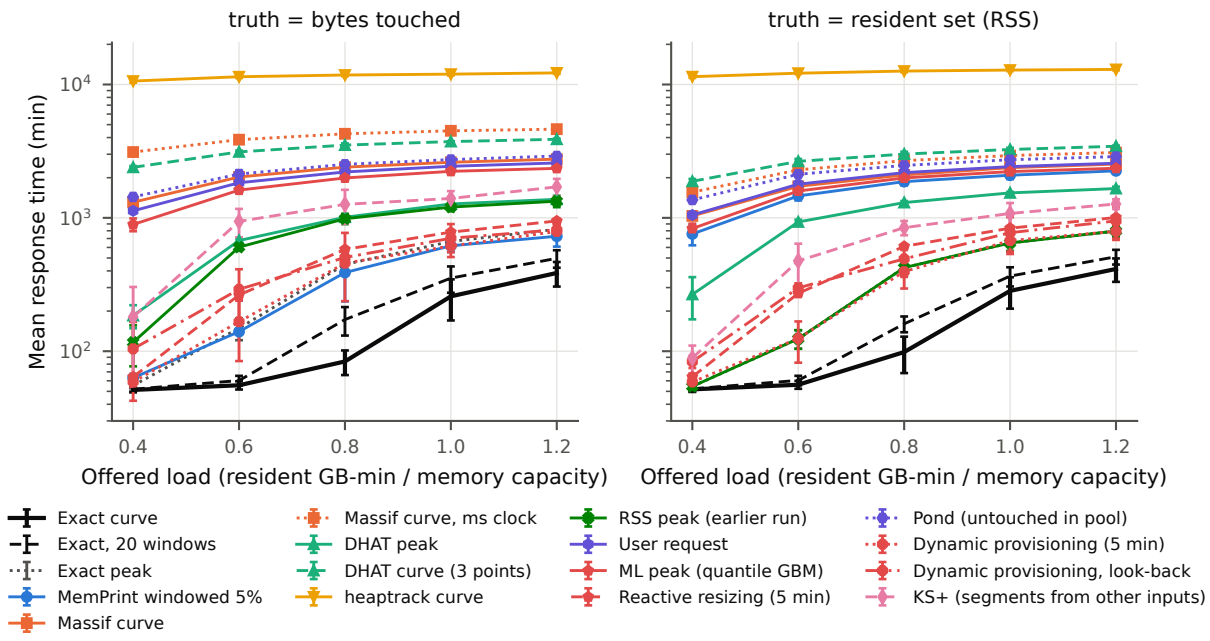


Figure 5: Mean response time against offered load, calibrated 90% margins (95% confidence intervals over three seeds, log scale). Left: the truth is bytes touched. Right: the truth is the RSS.

minutes, with 444 and 478 restarts and 25 and 27 jobs falling back to a static request per stream: behind the exact peak (448 and 423) and, under the bytes-touched truth, behind MemPrint's curve (389). The reactive oracle, which also resizes to the past five minutes but reserves its future resizes when the job starts, gives 580 and 612 minutes. Dynamic provisioning as simulated by its authors therefore bounds what a runtime monitor can reach; MemPrint's curve under the bytes-touched truth does better than the monitor a system could deploy, before the job starts and without restarts. Pond gives 2,519 minutes, worse than the request (2,208), and causes no kills. It reserves the whole request, so it cannot pack more jobs than the request does, and the share it places in the pool must fit in the pool even when the node's local memory is free. With a pool of 4 TB instead of 1 TB, Pond and the request give 302 and 307 minutes (one seed, mean over all jobs). Its model predicts 8% of the request as untouched against 38% in truth, at the 0.98 quantile, and still overpredicts for 11% of the jobs, because they come from users the model has not seen. Pond saves local DRAM, not total memory, which is the resource the job streams contend for.

KS+. KS+ gives 1,262 minutes under the bytes-touched truth and 844 under the RSS truth, better than the request and Pond but behind the exact peak, an earlier run's RSS and both resizing policies, with about 8,200 and 8,500 OOM kills per stream. For most programs its prediction of the largest input from the two smaller ones is close: the peak it first predicts is 0.86 to 1.18 times the true RSS peak for 19 of the 24 programs it can predict, and those jobs fail at most twice. Two programs cause nearly all the kills. The smaller inputs of atax and NPB MG barely rise above the runtime's baseline, so the regressions predict almost nothing for the largest input, and their jobs climb from the 100 MB floor in 20% steps, 36 to 38 times for atax and 20 to 23 times for MG. GCC's cc1 never fails but reserves 8.6 to 10.7 times its peak, because the two smaller source files predict a steep line. KS+ was designed for workflow tasks with hundreds of earlier executions; with three inputs per program, far apart in size, an input-size regression has little to extrapolate from,

Table 4: The margin each oracle needs to stay above the truth, as a share of its own peak (%), over the 25 programs at their largest inputs.

	Truth = bytes touched			Truth = RSS		
Oracle	Median	90th pct.	Max	Median	90th pct.	Max
MemPrint windowed 1%	0	14	136	6	69	150
MemPrint windowed 5%	0	6	40	6	57	112
MemPrint pages only	0	8	4112	7	79	4737
Massif curve	0	52	534	2	56	689
Massif pages-as-heap	0	28	62	0	30	63
Massif curve, ms clock	0	100	535	20	71	690
Massif pages-as-heap, ms clock	0	65	78	0	52	76
heaptrack curve	1	326	593	2	376	747
DHAT peak	0	6	546	2	24	706
DHAT curve (3 points)	36	90	580	53	99	739
DHAT hold-then-decay	21	90	580	39	99	739
gperftools peak	0	233	554	2	290	716
RSS peak (earlier run)	0	1	7	0	0	0

and its errors are the large errors that cost a scheduler most.

5.3 The RSS truth and the time axis

Under the RSS truth, the measured curves lose their advantage. MemPrint's windowed and pages-only curves give 1,869 to 2,516 minutes, worse than the DHAT peak (1,302 minutes) and the MemPrint 5% peak (959 minutes). The reason is the margin, not the shape. MemPrint's curve is on a clock of memory references, while the RSS truth is on wall-clock seconds, so a phase falls at a slightly different point of progress on the two curves, and MemPrint counts bytes rather than pages, which is half of miniVite's RSS. The margin that covers 90% of the programs is therefore 57% of the peak for MemPrint at 5%, against 6% under the bytes-touched truth (Table 4), and a curve with 57% headroom on its peak reserves more than a peak. A curve oracle is thus only as useful as its time axis is faithful to wall-clock time. Under the bytes-touched truth the MemPrint curves share the truth's clock of memory references, which favours them, and the RSS truth is the closer model of a real deployment. MemPrint does not record the native time of each snapshot, and Section 5.9 measures what placing its curve on the native clock does.

5.4 Margins

No margin policy rescues an estimator that is badly wrong for some programs (Figure 6). A small margin lets those programs' jobs be killed, and a margin large enough to cover them over-reserves every other job. With calibrated 100% margins, every heap profiler must cover GCC's cc1, whose peak it reads at 0.14 to 0.16, so each job reserves about 6.5 times its estimate and the response time rises to 15,000 to 17,000 minutes. Before the LLM engines joined the corpus, the MemPrint curves needed at most 7 to 19% under the bytes-touched truth; their late view of the mapped weights raises that to 40% at 5% watched, 136% at 1% and 41 times the estimate for pages only (llama2.c), so with calibrated 100% margins they give 1,299, 4,984 and 9,440 minutes. Taking each oracle at its best margin policy changes the order little (Figure 14 in the appendix): under the bytes-touched

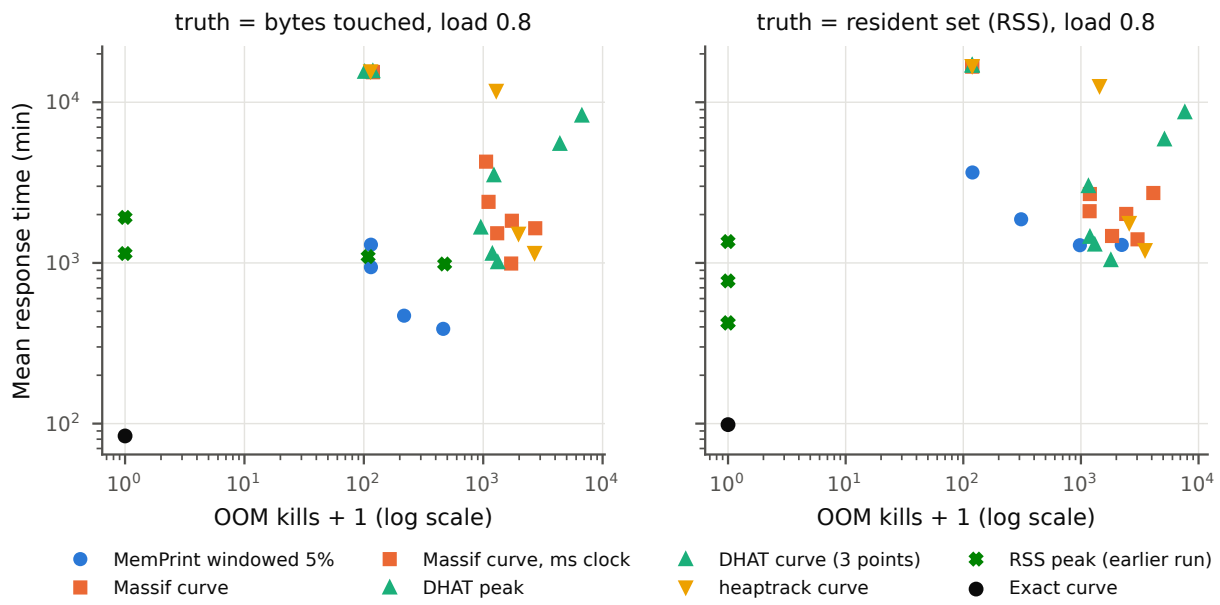


Figure 6: Mean response time against OOM kills at load 0.8, one marker per margin policy (10%, 30%, calibrated 90% and 100%). No margin gives a measured estimator both few kills and a short response time, except the MemPrint curves under the bytes-touched truth.

truth, MemPrint windowed at 5% gives 389 minutes, MemPrint pages only 469, Massif 991 (1,644 with its millisecond clock), DHAT 1,011 and heaptrack and gperftools 1,154 and 1,158.

5.5 Footprints predicted from source

The static route predicts nine of the programs (PolyBench 2mm, gemm and jacobi-2d at MEDIUM and atax at LARGE, HPCCG 64, LULESH 30 and NPB IS, CG and MG at class W), and we replay a stream made of those programs alone (Figure 7). Their footprints are flatter than those of the full corpus, so the exact peak comes close to the exact curve (273 against 194 minutes at load 0.8 under the bytes-touched truth). With calibrated 90% margins, the static predictions give 642 minutes, against 471 minutes for MemPrint windowed at 5%, 623 for heaptrack, 642 for Massif, 662 for the DHAT peak and 2,038 for the request, a 3.2× gain over the request without running the program. Two predictions cause most of the static oracle's 343 kills per stream: NPB MG is 7 to 9% short, and the margin calibrated on the other eight programs is 0%, which does not cover it. LULESH is over-predicted by 1.57 times, which wastes memory rather than causing kills. Under the RSS truth the static curve, like the windowed one, needs more headroom, and it gives 1,709 minutes against 1,399 for the DHAT peak.

5.6 LLM inference jobs

To see how the estimators fare on a kind of job whose memory is mostly read-only model weights, we replay a stream made of the LLM engines alone (Figure 8, Table 5). Each job takes the footprint and the estimates of one of the five traced configurations (llama2.c with the 15M, 42M and 110M models, and llama.cpp with 32 and 128 tokens), and each configuration's calibrated margin comes from the other four, as a site that has run other models or lengths of the same engines would set it.

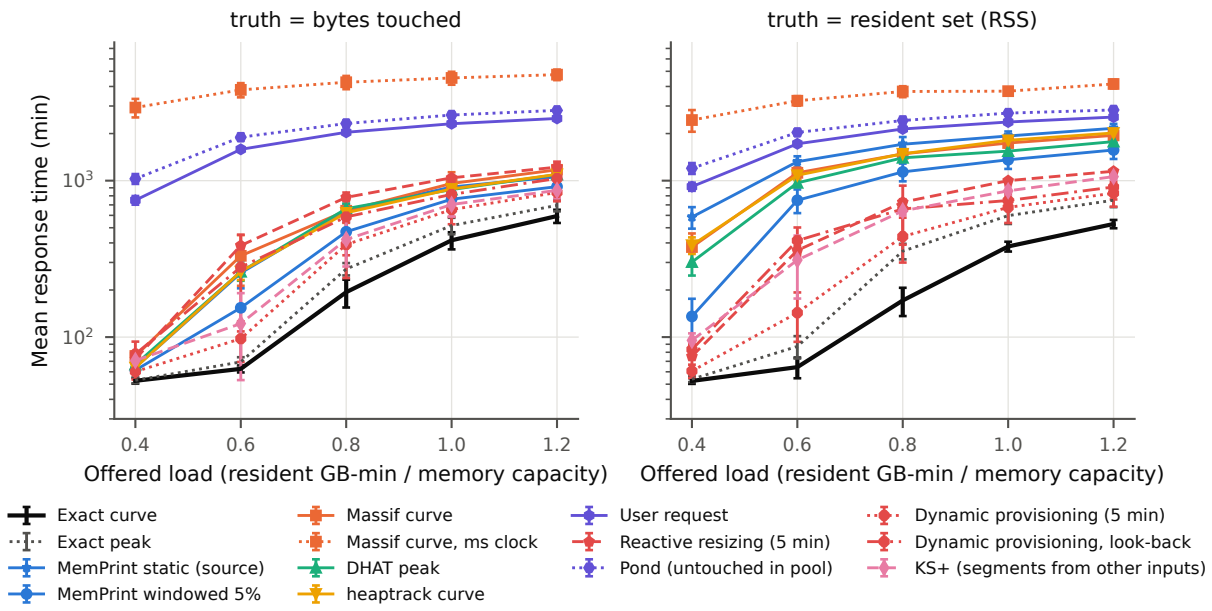


Figure 7: The nine programs the static route predicts (each at its largest input with a prediction): mean response time against offered load, calibrated 90% margins.

These footprints are flat, so the curve gains less over the peak than in the full corpus: at load 0.8 the exact curve gives 180 minutes, the exact peak 307 and the request 2,067 under the bytes-touched truth, a 1.7× gain of the curve over the peak against 5.3× for the full corpus. The best measured estimate is the RSS peak of an earlier run, 598 minutes with no kills under the bytes-touched truth and equal to the exact peak under the RSS truth, because the weights the RSS counts are the footprint. Reactive resizing gives 753 minutes and MemPrint's 5% peak 785 minutes, 602 kills per stream. Dynamic provisioning, which reads the next five minutes of the truth, gives 386 and 443 minutes with 140 and 200 restarts (528 and 694 for its look-back variant), Pond 2,379 and 2,450, and KS+, which predicts each configuration from the engine's other two, 737 and 547 minutes with about 1,540 and 800 kills.

Every heap profiler fails. They read llama2.c at 0.08 to 0.21 of its peak and llama.cpp at about 1.05, so the margin calibrated on the other configurations cannot cover llama2.c's jobs, which would need headroom of 3.8 to 10.6 times the estimate: with calibrated margins Massif, DHAT, heaptrack and gperftools give 8,747 to 9,344 minutes and 1,195 kills per stream, four times the request. With a fixed 10% margin, llama2.c's jobs are killed until the requeue factor has raised their reservation enough, 9,050 to 9,647 times per stream for DHAT, Massif and gperftools (15,661 for heaptrack), and their response of 613 to 823 minutes hides that waste. Massif with `-pages-as-heap` counts the file mapping and never under-reads, so it is never killed, but it over-reserves (2,650 minutes).

MemPrint's curves fail for a different reason. Its peaks are about right (0.86 to 1.00 of the true peak at 5% watched), but its curves reach the peak late (Section 4), while every job's footprint jumps to the weights in the first percent of its run. A curve reservation is therefore too small at the start of every job, and the configurations need 17 to 96% headroom at 5% watched; the calibrated margin covers some and not others, and the curves give 3,120 to 6,685 minutes with 610 to 1,049 kills per stream. Pages only, which never watches, needs headroom of 20 to 41 times its estimate for llama2.c. This is the same finding as under the RSS truth, from another cause: a curve is worth only as much as its timing, and MemPrint's windowed timing misses memory that a program maps

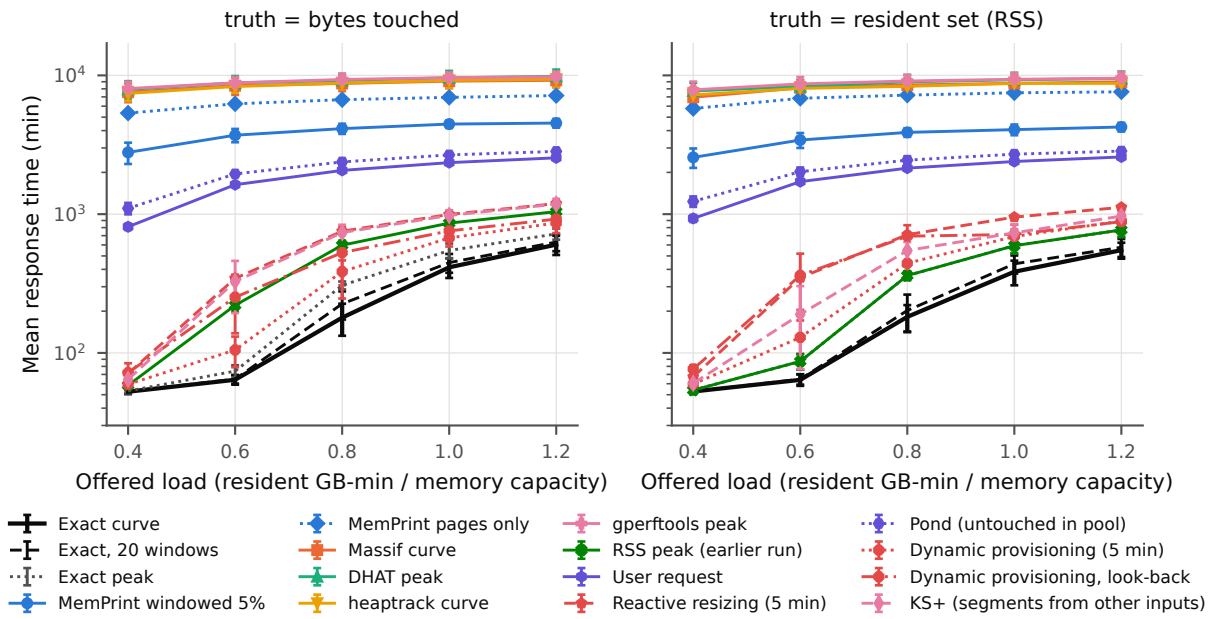


Figure 8: A stream of LLM inference jobs alone (llama2.c with three models, llama.cpp with two lengths): mean response time against offered load, calibrated 90% margins, each configuration's margin calibrated on the other four.

and reads before the first window. Section 5.8 adds file mappings to what the windowed mode tracks between windows.

5.7 A real trace

On the measured Lisa stream (Figure 9, Table 6), knowing a job's memory matters only once memory is scarce. At Lisa's own 103 GB and at 64 GB per node every policy completes the same 9.4 to 9.6 jobs per hour: most jobs take whole 16-core nodes and use a few GB, so the cores are the bottleneck, 98% busy. From 32 GB per node memory is contended, and the exact curve completes 8.6 jobs per hour against 6.6 for the exact peak (30% more, with a 40% shorter response time), and 18 to 31% more from 16 down to 4 GB. The peak and the request give the same results, because a whole-node job blocks its node whatever memory it reserves; what differs is whether it also holds pool memory it is not using. Reactive resizing comes between the peak and the curve. Dynamic provisioning restarts 775 to 2,129 jobs per stream at 32 GB per node and below and gains least of the curve policies; its look-back variant does better at 32 GB (7.3 jobs per hour) and worse at 16 GB (2.1).

The hybrid stream (Table 7), at 32 GB per node, replays the main results on the trace's mix of job sizes, runtimes and node counts. Under the bytes-touched truth, the exact curve gives 10,481 minutes and the exact peak 15,563; MemPrint's curves give 11,118 (pages only) and 12,954 minutes (windowed at 5%), ahead of the exact peak, both dynamic-provisioning variants (13,451 and 13,722), KS+ (19,920), Pond (19,710) and every heap profiler (21,440 to 59,035). Under the RSS truth, as in the main stream, MemPrint's curves on their clock of references lose that advantage (28,571 and 33,582 minutes), and reactive resizing and the look-back dynamic provisioning come next after the exact curves.

Table 5: The LLM stream at offered load 0.8, calibrated 90% margins (10% for the request, ML and reactive oracles, none for Pond and dynamic provisioning), mean of three seeds; columns as in Table 3.

	Truth = bytes touched					Truth = RSS				
Oracle	Resp. (min)	BSLD	Kills	Res. %	Used %	Resp. (min)	BSLD	Kills	Res. %	Used %
Exact curve	180	6.6	0	76	76	182	6.7	0	74	74
Exact, 20 windows	226	8.6	0	75	72	202	7.5	0	76	73
Exact peak	307	12.1	0	75	69	361	14.4	0	76	66
Dynamic provisioning (5 min)	386	15.8	140	72	65	443	18.3	200	71	63
Dynamic provisioning, look-back	528	22.2	260	68	62	694	29.3	356	62	56
RSS peak (earlier run)	598	24.9	0	69	57	361	14.4	0	76	66
KS+ (segments from other inputs)	737	31.1	1542	64	49	547	22.8	797	69	56
Reactive resizing (5 min)	753	31.9	10	65	54	712	29.9	9	66	54
MemPrint windowed 5%, peak	785	32.9	602	68	51	913	38.4	602	68	46
ML peak (quantile GBM)	1836	79.1	115	61	36	1904	82.1	115	61	34
User request	2067	89.3	0	58	33	2146	92.7	0	57	31
Pond (untouched in pool)	2379	102.8	0	52	30	2450	105.9	0	53	28
Massif pages-as-heap	2650	114.7	0	47	26	2186	94.5	0	52	30
Massif pages-as-heap, ms clock	2955	127.9	0	45	24	2495	107.9	0	48	26
MemPrint windowed 1%	3120	135.7	1049	46	24	3083	134.0	1046	47	23
MemPrint windowed 5%	4134	180.6	1034	43	19	3884	169.6	1033	44	19
MemPrint pages only	6685	293.2	610	29	9	7208	316.3	1220	30	9
Massif curve	8747	381.2	1195	32	9	8528	372.3	1195	32	10
heaptrack curve	8794	383.2	1195	32	9	8333	363.2	1195	32	10
Massif curve, ms clock	8869	386.7	1195	32	9	8630	376.5	1195	32	10
DHAT curve (3 points)	9103	397.6	1195	31	9	8776	383.4	1195	32	9
DHAT peak	9198	400.7	1195	31	9	8986	391.7	1195	32	9
gperftools peak	9344	408.0	1195	31	9	9108	397.5	1195	32	9
DHAT hold-then-decay	9561	417.3	1195	31	8	9251	404.3	1195	32	9

5.8 Tracking file mappings between windows

Between windows, MemPrint's windowed mode tracks the program's heap blocks and anonymous mappings and reads their resident pages from `/proc/self/pagemap` at every snapshot, but a file mapping was not a block, so its pages were seen only through the samples a window took. We added the program's file mappings to the tracked memory (`-file_maps`, now the default). The tool catches every `mmap` of a file without `PROT_EXEC` at the system call, trims the mapping on `munmap` or when a later mapping replaces part of it, and leaves out mappings inside a loaded image's address range, which are the loader's mappings of a library whose writable sections are already counted. At every snapshot the resident bytes of the remaining file mappings are added to the estimate, as those of a fresh heap block are, and the windows' samples that fall in them are not counted a second time. A test program that maps a 4 MB file and reads it before its first window is estimated at 4.1 MB at its peak with the option and at 0.14 MB without it.

We repeated MemPrint's windowed and pages-only runs at every program's largest input and at every LLM input with the new tool, and replayed both streams (Table 8). Only the LLM engines change materially: their peaks become 1.00 for every mode (llama2.c pages only: from 0.02), and the curves reach 90% of the peak at the same point of the run as the truth. Perl and SQLite, the only other programs whose peak moves, change by 1%. The margin that covers 90% of the programs falls to 1% at 5% watched, 4% at 1% and 4% for pages only (from 6, 14 and 8%), and in the LLM stream to 1% for every mode (from 92%, 137% and 37 times the estimate). On the main stream under the bytes-touched truth, all three MemPrint curves now beat the exact peak (448 minutes): windowed at 5% gives 327 minutes (389 before), at 1% 405 (765) and pages only 411 (469), and windowed at 5% needs 160 GB per node for the target service level, as the exact peak does, instead of 192 GB. On the LLM stream the curves give 365 to 386 minutes instead of 3,120 to 6,685, better than the earlier run's RSS peak (598) and close to the exact peak (307); the

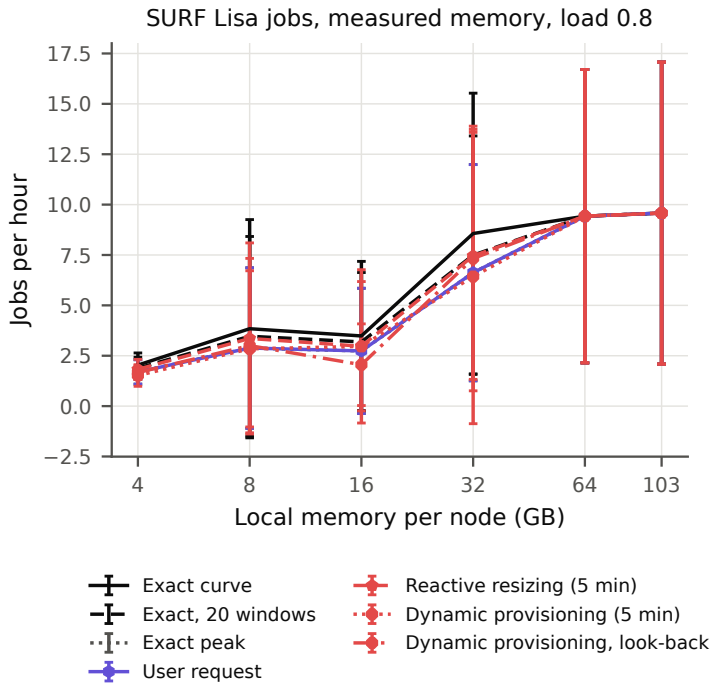


Figure 9: SURF Lisa jobs with their measured memory on a Lisa-shaped cluster (32 nodes of 16 cores, a pool of a quarter of the local memory), offered load 0.8: jobs per hour against local memory per node (mean and 95% interval over three seeds). At Lisa's own 103 GB per node the cores limit every policy alike.

remaining 597 kills per stream come from llama2.c with the 15M model, which needs 1.2 to 1.4% of headroom, more than the margin calibrated on the other four configurations (under 0.5%). The option does not change the time axis, so under the RSS truth the curves still need their wall-clock headroom: 1,843 to 1,873 minutes on the main stream and 930 to 948 on the LLM stream, against 423 and 361 for the RSS peak.

5.9 The time axis and the level

Under the RSS truth, MemPrint's curves needed about 57% headroom, and we attributed it to their clock. Three changes close most of that gap (Table 9, Figure 10).

The native clock. The time under Pin is not the native time, because Pin slows a program's phases unevenly: GAP's graph construction, which holds its peak, takes a fifth of the native run but a twentieth of the run under Pin. We therefore measure the native clock: one more native run under `perf stat -I 5` counts the user-level loads and stores every 5 ms, which gives the native time at which each share of the run's memory references has executed, and a MemPrint sample taken after a share f of the references is placed at that time. No instrumentation is involved, and the run costs what the program costs. On the native clock the margin that covers 90% of the programs falls from 57% to 24%, and the response time from 1,873 to 694 minutes.

The level. The RSS counts memory MemPrint does not: code, the allocator's chunk headers and the memory it keeps after a free. A hybrid oracle keeps the shape of MemPrint's curve on the native clock and scales it so its peak is the peak of an earlier run's RSS, so the curve knows when and the RSS how much. Its margin falls to 17%, and it gives 609 minutes at 5% watched and 569 for pages only. With a fixed 10% margin it gives 415 to 417 minutes against 423 for the exact peak,

Table 6: The measured Lisa stream at offered load 0.8: jobs per hour and mean response time for each local memory per node (mean of three seeds; dynamic provisioning's look-back variant at 4 GB, loads 0.6 to 1.2: two seeds). The load is set against each cluster's memory, so numbers compare across rows, not across columns.

	Jobs per hour						Mean response (min)					
Local GB per node	103	64	32	16	8	4	103	64	32	16	8	4
Exact curve	9.6	9.4	8.6	3.5	3.8	2.0	11,790	10,755	9,735	36,491	36,939	14,073
Exact, 20 windows	9.6	9.4	7.5	3.2	3.5	1.9	11,790	10,755	12,264	46,250	46,383	20,655
Exact peak	9.6	9.4	6.6	2.7	2.9	1.7	11,790	10,755	16,080	59,143	60,184	28,962
User request	9.6	9.4	6.6	2.7	2.9	1.7	11,790	10,755	16,080	59,143	60,184	28,962
Reactive resizing (5 min)	9.6	9.4	7.5	3.0	3.4	1.8	11,790	10,755	11,705	46,413	44,847	18,800
Dynamic provisioning (5 min)	9.6	9.4	6.4	3.0	2.8	1.5	11,787	10,755	20,327	59,675	51,699	26,652
Dynamic provisioning, look-back	9.6	9.4	7.3	2.1	3.0	1.7	11,787	10,755	14,802	58,906	56,997	14,948

with 313 to 316 kills per stream.

The code. The RSS truth scales a program's whole RSS above its smallest sample with the job's size, so the code of the program and of its libraries, which does not grow with the job's data, is scaled as if it did. A second native run with the sampler reading each executable mapping's resident bytes from `smaps` measures the code: 1 to 27 MB, 10.4 of miniVite's 30.6 MB, 19.4 of cc1's 134 and 27 of llama2.c's 514. With the code left out of the truth (and of the earlier run's RSS), the hybrid with pages only gives 422 minutes against 429 for the exact peak. These are the first measured curves that keep up with the exact peak under a resident-set truth; plain MemPrint gives 1,920 minutes there.

What remains. No curve beats the peak under the RSS truths, as the exact curve does (89 to 99 minutes). Even on the native clock, the true footprint of GAP's construction holds its peak for one or two twentieths of the run, while the RSS holds its 76 MB for four, and Lua's RSS stays at 62 MB after its live data has fallen to 17 MB: the allocator keeps freed memory resident, which an estimate of live bytes cannot see. A curve of the RSS itself, taken from an earlier run, would see it, and is the next oracle to test.

On the LLM stream under the RSS truth, the hybrids give 400 to 417 minutes against 948 for plain MemPrint and 361 for the exact peak. In the memory sweep, MemPrint's native-clock curves meet the service level with 192 GB per node under both RSS truths (7.0 TB), instead of 256 GB on their clock of references; the exact peak needs 160 GB.

5.10 Memory needed for the same service

The response-time gains translate into memory a site would have to buy (Figure 11). At a fixed arrival rate, the exact curve meets the target with 128 GB per node (5.0 TB in all). Under the bytes-touched truth, the exact peak needs 160 GB per node (6.0 TB, 20% more), MemPrint's windowed and pages-only curves and reactive resizing 192 GB (7.0 TB, 40% more), DHAT, an earlier run's RSS and the ML peak 256 GB (9.0 TB, 80% more), and Massif's curve and the request 320 GB (11.0 TB, 120% more). The DHAT curve needs 384 GB per node (13.0 TB) and Massif's curve on its millisecond clock 512 GB (17.0 TB), and gperftools and heaptrack do not meet the target even at 512 GB per node. With the LLM engines in the corpus, MemPrint's curves need one step more than before (192 instead of 160 GB per node), and gperftools and heaptrack fall from 256 GB to never, through their calibrated margins. Under the RSS truth, the exact peak and the RSS of an earlier run need 6.0 TB, reactive resizing 7.0 TB, MemPrint windowed at 5%, DHAT, Massif, the ML peak and the request 9.0 TB, and MemPrint pages only 11.0 TB, which repeats the finding that

Table 7: The hybrid Lisa stream (the trace's jobs with corpus footprint shapes) at 32 GB per node and offered load 0.8, calibrated 90% margins as in Table 3. Fails counts OOM kills and dynamic provisioning's restarts.

	Truth = bytes touched			Truth = RSS		
Oracle	Jobs/h	Resp. (min)	Fails	Jobs/h	Resp. (min)	Fails
Exact curve	6.9	10,481	0	6.8	10,590	0
MemPrint pages only	5.5	11,118	1893	3.9	33,582	1496
Exact, 20 windows	6.7	11,585	0	6.7	12,289	0
Reactive resizing (5 min)	6.7	11,740	2	6.5	12,983	4
MemPrint windowed 5%	6.3	12,954	453	4.2	28,571	306
Dynamic provisioning, look-back	6.6	13,451	251	6.6	13,749	282
Dynamic provisioning (5 min)	6.6	13,722	405	6.2	14,945	515
User request	6.3	15,563	0	6.3	16,012	0
Exact peak	6.3	15,563	0	6.3	16,012	0
Pond (untouched in pool)	6.0	19,710	0	6.0	19,677	0
KS+ (segments from other inputs)	4.8	19,920	6136	3.3	17,157	6411
RSS peak (earlier run)	5.8	20,623	474	6.3	16,012	0
DHAT peak	5.2	21,440	1323	4.9	24,514	1310
ML peak (quantile GBM)	4.1	21,892	212	4.0	21,682	212
Massif curve	3.9	31,021	1092	3.8	30,744	1188
DHAT curve (3 points)	3.1	45,306	1197	3.1	38,652	1149
Massif curve, ms clock	3.2	46,978	1040	3.6	34,698	1190
heaptrack curve	4.1	47,648	1285	3.8	45,731	1440
gperftools peak	3.3	59,035	1644	3.6	57,569	1644

a curve on a reference clock loses its advantage against a wall-clock truth. Dynamic provisioning needs 7.0 TB under the bytes-touched truth and 6.0 TB under the RSS truth, like the exact peak; its look-back variant needs 7.0 and 9.0 TB. Pond needs 512 GB per node (17.0 TB) under both truths, because the pool, not local memory, is what its placement fills. KS+ needs 256 GB per node (9.0 TB) under both truths, as DHAT and the ML peak do: its retries keep a few programs' jobs failing, and cc1's over-reservation holds memory the others could use.

5.11 What each estimator's cost buys

Figure 12 sets each estimator's cost against what it buys. Under the bytes-touched truth, the full Pin trace costs 227 times the native run and gives a 26.4× gain over the request as a curve, or 4.9× as a peak. MemPrint windowed at 5% costs 11 times the native run and gives 5.7×, about a fifth of the exact curve's gain at a twentieth of its cost; pages only and windowed at 1% cost 6 and 9 times and give 4.7× and 4.3×. heaptrack and gperftools cost under 1.5× and give 1.9×, DHAT costs 24× and gives 2.2×, and Massif's curve costs 5.7× and gives 2.2× (1.3× with its millisecond clock). An earlier run's RSS costs nothing extra and gives 2.2× under the bytes-touched truth, and under the RSS truth it is the truth's peak. Under the RSS truth the measured curves give 1.6 to 1.7× and the heap profilers' peaks 1.8 to 2.1×, so the profiling cost buys little until the curve's time axis is fixed.

6 DHAT curves

DHAT records the heap at its global peak (t-gmax), the time of that peak, and the heap at exit, but no time series [20]. A scheduler can use those three points in two ways besides the peak. The *DHAT curve* rises linearly from zero to the peak at its recorded time and falls linearly to the exit

Table 8: MemPrint's oracles before and after tracking file mappings between windows (`-file_maps`): mean response time (minutes) at load 0.8, calibrated 90% margins, mean of three seeds. Only MemPrint's windowed runs were repeated; every other oracle is unchanged.

	Main stream (25 programs)				LLM stream			
	Bytes touched		RSS		Bytes touched		RSS	
Oracle	before	after	before	after	before	after	before	after
MemPrint windowed 5%	389	327	1869	1873	4134	386	3884	948
MemPrint windowed 1%	765	405	2340	1855	3120	370	3083	930
MemPrint pages only	469	411	2516	1843	6685	365	7208	939
MemPrint windowed 5%, peak	818	817	959	958	785	601	913	986

Table 9: Mean response time (minutes) at load 0.8 under three truths, the main stream, calibrated 90% margins. MemPrint's runs track file mappings (Section 5.8). Under the RSS truths, an earlier run's RSS peak is the truth's own peak, and equals the exact peak.

Oracle	Truth = bytes touched	Truth = RSS	Truth = RSS without code
Exact curve	84	99	89
Exact peak	448	423	429
RSS peak (earlier run)	985	423	429
MemPrint windowed 5%	327	1,873	1,920
MemPrint 5%, native clock	1,549	694	564
MemPrint pages, native clock	1,561	783	667
MemPrint 5%, native clock, RSS level	2,053	609	469
MemPrint pages, native clock, RSS level	2,028	569	422
DHAT peak	1,011	1,302	1,225
Massif curve	2,399	2,099	2,264
heaptrack curve	11,778	12,610	12,410
gperftools peak	10,476	11,181	11,106
User request	2,208	2,183	2,202

value. The *hold-then-decay* curve holds the peak from the start to its recorded time, which is safe before the peak, and falls linearly afterwards. Only "never above the peak" is guaranteed by what DHAT records, and that is the DHAT peak oracle.

Both DHAT curves do worse than the DHAT peak under every margin policy and both truths, except with calibrated 100% margins, where all three reserve for GCC's cc1 and give about the same (15,000 to 17,000 minutes; Figure 13). At load 0.8 with calibrated 90% margins, the DHAT curve gives 3,512 minutes and the hold-then-decay curve 4,161 minutes, against 1,011 minutes for the DHAT peak, under the bytes-touched truth, and 3,008, 3,943 and 1,302 minutes under the RSS truth. With fixed margins the curves are worse still (8,263 and 10,648 minutes at 10%), because they fall below the truth wherever a program has more than one phase, and their jobs are killed 5,751 to 7,642 times per stream. The median program needs 36% headroom on the DHAT curve (21% on hold-then-decay) and the 90th percentile needs 90% (Table 4). The average of the DHAT curve over the run is 0.81 times the true mean footprint at the median over the 71 (program, input) pairs, so it under-reserves on average as well as in places. DHAT's three points therefore do not substitute for a measured curve. The information that makes a curve useful to a scheduler is the

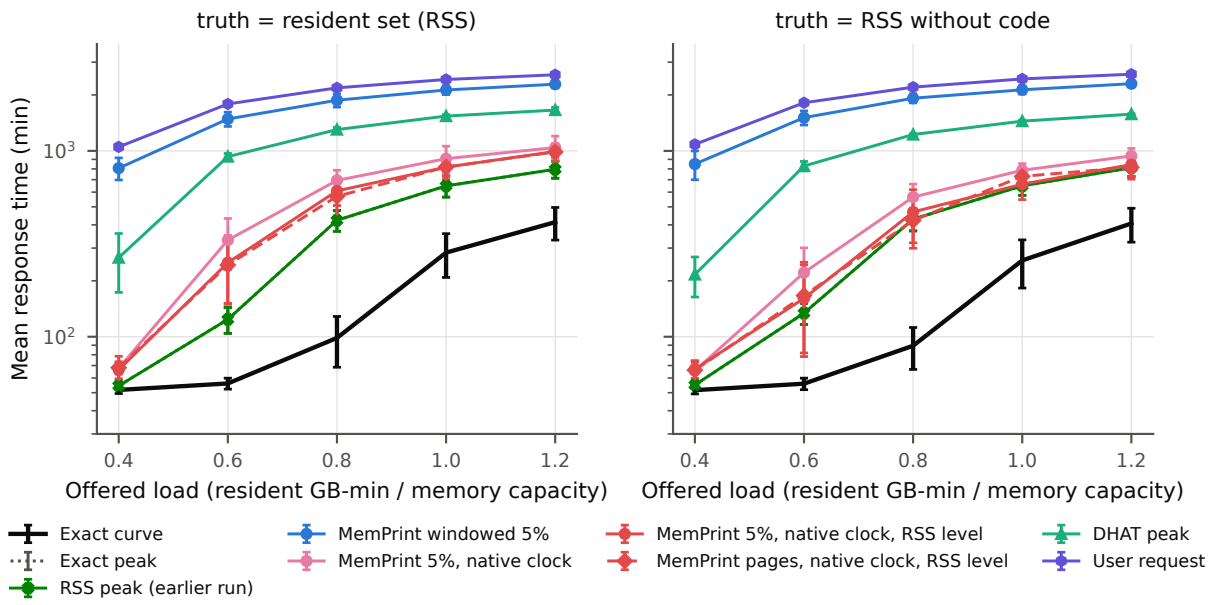


Figure 10: Mean response time against offered load under the RSS truth and the RSS truth without code, calibrated 90% margins: MemPrint's curve on its clock of references and on the native clock, and scaled to an earlier run's RSS peak. The exact peak lies under the RSS peak of an earlier run.

timing of every phase, and DHAT records the timing of one.

7 Limitations

The results rest on a model of a cluster and of its jobs, and each simplification limits what they show.

The published baselines. Pond's artifact holds none of its policies, so our Pond follows the paper's description; it leaves out the model that places latency-insensitive virtual machines entirely in the pool, which needs hardware performance counters our jobs do not have. Our dynamic provisioning fixes a job's runtime stretch at admission and charges nothing per resize, both in its favour, and its resizes read the coming interval as the authors' simulator does. KS+ learns from two smaller inputs per program, where its authors had hundreds of executions per task.

The Lisa trace. Its jobs are those that had their nodes to themselves (88,767 of 267,384 jobs with metrics), which leaves out the jobs that shared nodes, mostly small ones. Active memory counts the page cache the job keeps active and not the memory it has mapped and not touched, and the node's idle level is taken as its 1st percentile over the whole trace. The trace records no memory request, so the request is Slurm's default for the simulated node.

Scaled footprints. The corpus programs peak at 15 to 450 MB, and the simulated jobs at tens to hundreds of GB. Scaling a measured curve to a job's size keeps its shape and every estimator's ratio to the truth, but it assumes the ratio does not change with the input. The corpus shows that it often does at small inputs: MemPrint windowed at 5% reads 0.61 to 0.92 times the peak of the

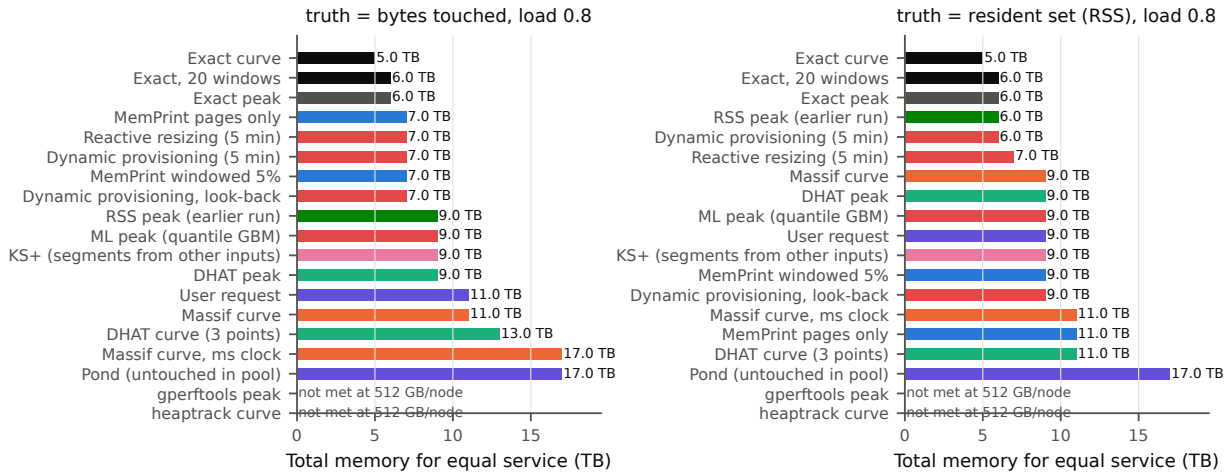


Figure 11: The total memory (32 nodes of local DRAM plus the 1 TB pool) each oracle needs to keep the mean bounded slowdown within 1.5 times that of the exact curve at 128 GB per node, at a fixed arrival rate (load 0.8 on the 128 GB cluster). Local memory per node was swept over 48, 64, 96, 128, 160, 192, 256, 320, 384 and 512 GB.

PolyBench kernels at SMALL, and 1.00 at LARGE. We use the largest input of every program to keep this effect small.

Time axes. Every curve is put on the job's progress by linear rescaling of its own clock. The true footprint of a job over wall-clock time is known only for the RSS truth, so the bytes-touched truth favours the MemPrint curves, which share its clock of memory references (Section 5).

The job stream. The main stream is synthetic, with memory the contended resource by design, and the used-to-requested ratio comes from the LANL CM5 log of 1994 to 1996, the only archive log with both fields that we found. A real trace with requested and used memory per job would test whether the effect holds at the memory pressure of a production system.

Scheduler and machine. Jobs use one node, the slowdown is linear in the remote share of the footprint with one β for every job, the pool has no bandwidth contention, and only FCFS with EASY backfilling was tested. The OOM model requeues a killed job with 1.5 times its reservation, and a batch system that killed a job only once, or that swapped instead of killing, would change the kill counts.

Costs. The costs were measured with up to 15 runs sharing the node, so they are slowdowns under load, and Pin's fixed start-up cost weighs heavily on runs of a few seconds. Longer jobs would amortise it.

LLM inference. The LLM jobs are single completions of 32 to 128 tokens on the CPU, scaled like every other program. A serving engine that runs for hours fills its KV cache over many requests, and a GPU engine keeps its weights in device memory, so neither shape is covered here. llama.cpp's 360M model could not be traced, so the corpus has llama.cpp at two inputs of one model.

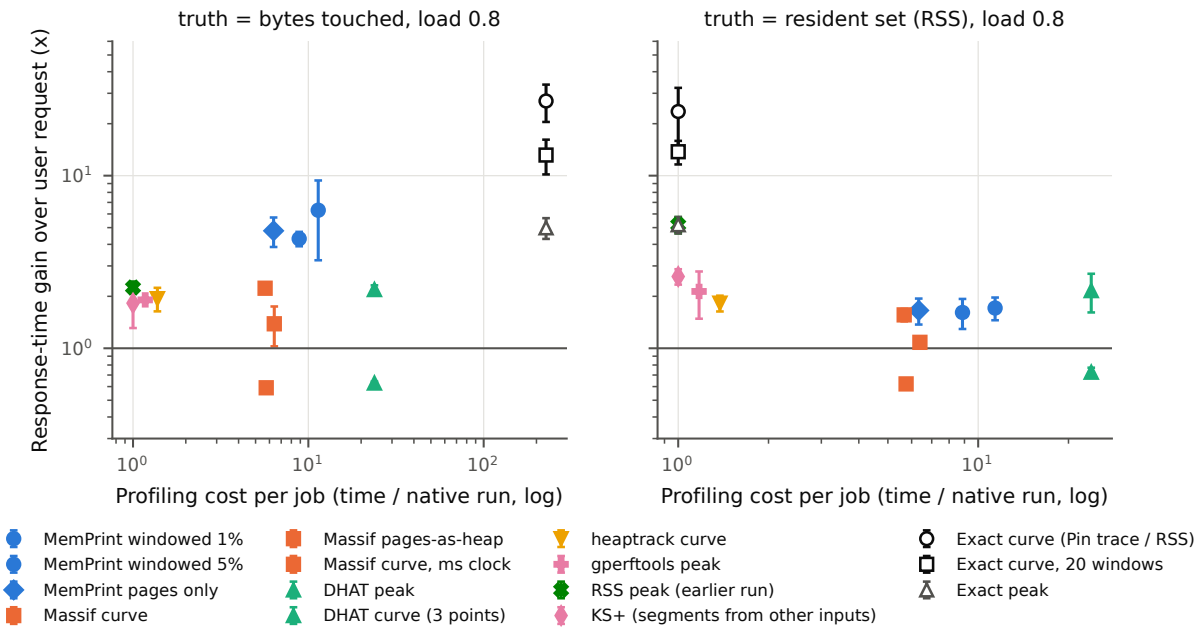


Figure 12: Profiling cost per job (median time over the native run, at the largest inputs) against the gain in mean response time over the user's request, at load 0.8, each oracle at its best margin policy. The exact curve's cost is the full Pin trace under the bytes-touched truth and an RSS sampler under the RSS truth.

The static route. The static predictions cover 9 of the 25 programs. They predict a whole-run footprint for C++ programs, so a program that frees and reallocates its data, such as LULESH, is over-predicted, and the interpreter takes up to 5.5 hours for one input of NPB CG.

8 Conclusion

An exact footprint is worth most to a scheduler on disaggregated memory as a curve over the job's run, and the gain comes from letting jobs with peaks at different times share memory. On our corpus and streams, the exact curve gives a 26.4× shorter mean response time than the user's request and needs 5.0 TB where the request needs 11.0 TB. Heap profilers get most peaks right, but their large errors on a few programs, which come from memory that is mapped outside the heap or allocated and never touched, cannot be covered by any margin without over-reserving every other job; LLM engines, whose weights are a file mapping, add a third such program, which pushes the calibrated margins of heaptrack and gperftools past twice their estimates. Among measured estimators, MemPrint windowed at 5% comes closest to the exact curve, with the smallest margin (6% for 90% of programs) and 40% more memory than the exact curve for the same service, at 11 times the native run time. On LLM jobs alone it failed, because its curve learned of the weights only in its windows, while the footprint reaches its peak in the first percent of the run; tracking the program's file mappings between windows, as its heap blocks are tracked, removed that failure and made every MemPrint curve beat the exact peak on the full corpus. DHAT's three recorded points do not make a usable curve. Of three published policies, the one that comes closest to the exact peak, dynamic provisioning, does so by reading the coming minutes of a job's use from its trace, which a running system cannot do, and by restarting the jobs whose growth no longer fits (looking back instead, it falls behind the exact peak); Pond saves local DRAM but not total memory, and

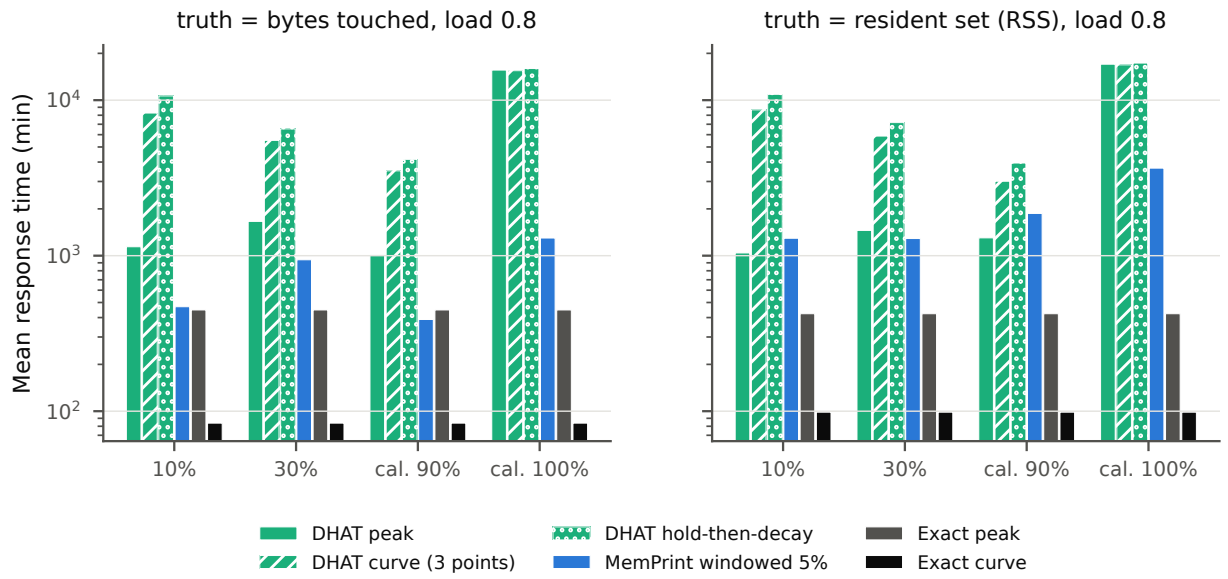


Figure 13: DHAT's peak and the two curves built from its three recorded points, against MemPrint windowed at 5% and the exact peak and curve, at load 0.8 under each margin policy (log scale).

KS+'s regressions on input size cannot extrapolate from inputs far smaller than the job's. Knowing a job's footprint before it starts, which no published policy does, remains the missing piece. Two results set the direction of further work. First, under the RSS truth every measured curve lost its advantage; placing MemPrint's curve on the native clock through hardware counts of memory references, at the level of an earlier run's RSS, brings it to the exact peak, but beating the peak there needs an estimate of the memory an allocator keeps after a free. Second, the static route predicts footprints without running the program, but its errors on programs that free and reallocate memory, and its cost on large inputs, limit it today to programs whose footprint is set by their input sizes.

A Additional figures

B Results at every offered load

The results of Section 5 are given at offered load 0.8. This appendix gives each of them at loads 0.4, 0.6, 1.0 and 1.2 as well, from the same runs (three seeds per point): the main stream (Table 10 to Table 13), the margins (Figure 16), what each estimator's cost buys (Figure 17), DHAT's curves (Figure 18), the LLM stream (Table 14 to Table 17), the file mappings (Table 18 to Table 21), the time axis and the level (Table 22 to Table 25), and the memory needed for the same service, for which the memory sweep was rerun at every load (Table 26, Figure 19).

Where load 0.8 sits. At load 0.4 no oracle that does not over-reserve leaves jobs waiting, so the exact curve, the exact peak, MemPrint's curves under the bytes-touched truth, reactive resizing and dynamic provisioning all give 51 to 71 minutes, close to the runtimes. Only the oracles that over-reserve (the request, Pond, the ML peak, the profilers with large calibrated margins) or are killed repeatedly (DHAT, and the RSS peak under the bytes-touched truth) stand apart. From

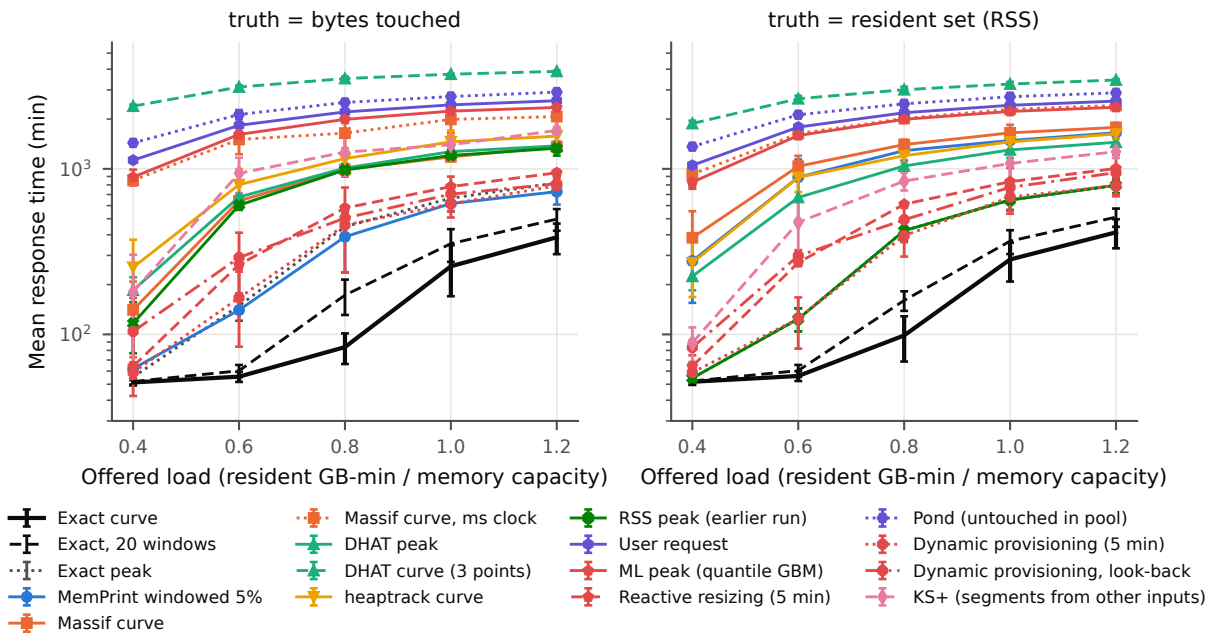


Figure 14: Mean response time against offered load, each oracle at its best margin policy at each load.

load 0.6 memory is contended, and the gain of the exact curve over the exact peak is 2.7× at 0.6, 5.3× at 0.8, 2.6× at 1.0 and 2.1× at 1.2 under the bytes-touched truth (2.2, 4.3, 2.3 and 1.9× under the RSS truth). Load 0.8 is where the curve gains most: there the curve still keeps up with arrivals (84 minutes) while the peak no longer does; above it, the curve saturates as well and both queue. The gain over the request falls from 22 to 33× at loads up to 0.8 to 9.4× at 1.0 and 6.7× at 1.2, because the request saturates first and gains little from more load.

What holds at every load. The order of the oracles hardly changes with load. Under the bytes-touched truth, MemPrint windowed at 5% beats the exact peak at every contended load (140, 389, 617 and 728 minutes against 149, 448, 665 and 822 at loads 0.6 to 1.2), and it and dynamic provisioning are the two best oracles other than the exact ones at every load, followed by MemPrint pages only and reactive resizing in either order. At load 0.4, where the exact peak needs no more than the curve, the exact peak (55 minutes) beats every estimate. Under the RSS truth, the RSS peak of an earlier run and dynamic provisioning are the two best at every load, and MemPrint's curves on their clock of references fall behind the exact peak at every load, as at load 0.8 (Section 5.3). On the LLM stream, the RSS peak of an earlier run and dynamic provisioning lead at every load, and the heap profilers fail at every load. On the programs the static route predicts, dynamic provisioning leads at every load and MemPrint windowed at 5% is second under the bytes-touched truth. Pond is slower than the request at every load and on every stream, because it reserves the whole request and holds part of it in the pool. KS+ is between the resizing policies and the request at every load on the main stream, and its number of kills does not depend on the load (8,190 to 8,550 per stream), because they come from the same two programs' predictions.

Memory needed for the same service. The memory sweep was rerun at every load, over 48 to 512 GB of local memory per node (Table 26). The target is set at each load as 1.5 times the exact curve's bounded slowdown at 128 GB per node, which is 1.5 to 1.8 at loads 0.4 and 0.6, 3.6

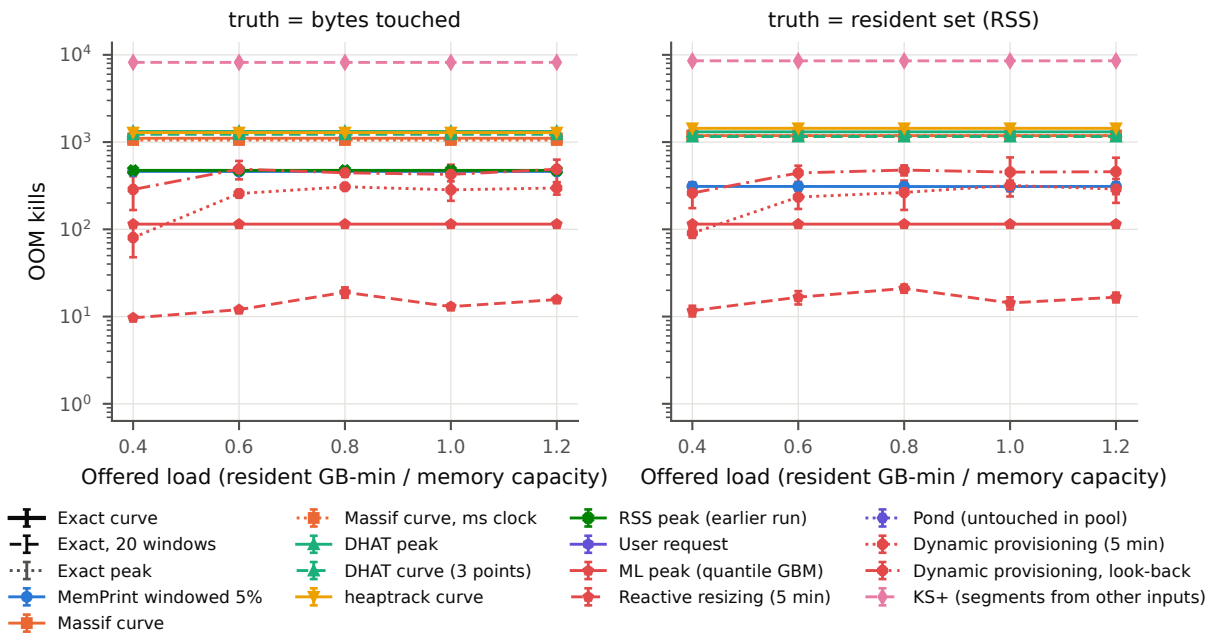


Figure 15: OOM kills per stream of 3,000 jobs against offered load, calibrated 90% margins.

to 4.6 at 0.8, and 15 to 25 at loads 1.0 and 1.2, where even the exact curve queues. The ranking hardly moves with load. The exact curve needs 4.0 TB at load 0.4 and 5.0 TB at every other load, and the exact peak one step more (5.0 to 6.0 TB). Under the bytes-touched truth, MemPrint's curves and dynamic provisioning need the same as the exact peak, or one step more at load 0.8 (6.0 to 7.0 TB), reactive resizing 6.0 to 7.0 TB, an earlier run's RSS and DHAT 6.0 to 9.0 TB, the ML peak 9.0 TB and the request 9.0 to 11.0 TB at every load. KS+ needs 6.0 TB at load 0.4 and 9.0 TB at the others. Pond needs 11.0 to 17.0 TB, more than the request at every load, and gperftools and heaptrack do not meet the target at 512 GB per node at any load. Under the RSS truth, the RSS of an earlier run and dynamic provisioning need what the exact peak needs (5.0 to 6.0 TB) at every load, while MemPrint's curves on their clock of references need 7.0 to 11.0 TB. The reruns of Section 5.8 and Section 5.9 keep their findings at every load (Table 27 and Table 28): with file mappings tracked, MemPrint's windowed curve needs what the exact peak needs under the bytes-touched truth at every load, and on the native clock its curves need 5.0 to 7.0 TB under both RSS truths, against 7.0 to 9.0 TB on the clock of references.

C Sensitivity to the simulator's assumptions

We varied each of the simulator's assumptions alone, at offered load 0.8. The remote-memory slowdown β changes every oracle's response time, from 58 to 390 minutes for the exact curve between $\beta = 0.1$ and 2, but not their order (Table 29): under the bytes-touched truth MemPrint windowed at 5% beats the exact peak at every β (236 against 274 minutes at 0.1, 825 against 1,162 at 2), and the request and Pond stay last. The pool decides how much memory is contended (Table 30): with 4 TB every oracle but the request, Pond and KS+ comes within a few minutes of the runtimes, and with 512 GB the differences grow (the exact curve 565 minutes, the exact peak 1,499, the request 4,441). The update interval matters most for the policies that resize a running job (Table 31): dynamic provisioning gives 157 minutes with an update every minute and 1,330

Table 10: Every oracle at offered load 0.4, as Table 3.

	Truth = bytes touched					Truth = RSS				
Oracle	Resp. (min)	BSLD	Kills	Res. %	Used %	Resp. (min)	BSLD	Kills	Res. %	Used %
Exact curve	51	1.0	0	44	44	52	1.1	0	44	44
Exact, 20 windows	52	1.1	0	49	44	52	1.1	0	49	44
Exact peak	55	1.2	0	58	44	54	1.2	0	56	45
Dynamic provisioning (5 min)	59	1.3	80	53	45	59	1.3	90	54	46
MemPrint windowed 5%	63	1.5	461	57	44	757	31.5	310	55	31
Reactive resizing (5 min)	64	1.6	10	57	44	64	1.6	12	56	44
MemPrint pages only	65	1.5	1901	58	43	1404	60.0	1495	50	25
MemPrint windowed 1%	71	1.7	1116	58	39	1199	50.9	620	51	26
MemPrint windowed 5%, peak	79	2.0	379	66	45	114	3.4	492	63	42
Dynamic provisioning, look-back	104	3.3	286	54	46	83	2.3	261	53	45
RSS peak (earlier run)	117	3.6	474	64	41	54	1.2	0	56	45
KS+ (segments from other inputs)	181	6.5	8198	60	38	89	2.4	8548	60	43
DHAT peak	185	6.6	1323	61	39	266	10.1	1310	60	37
ML peak (quantile GBM)	892	37.6	115	61	30	831	34.8	115	61	31
User request	1128	48.1	0	58	28	1051	44.6	0	58	29
Massif curve	1309	55.9	1107	51	25	1034	43.7	1184	52	28
Pond (untouched in pool)	1435	61.4	0	53	25	1362	58.1	0	53	27
DHAT curve (3 points)	2392	103.5	1228	48	22	1879	81.1	1157	48	23
DHAT hold-then-decay	3042	131.7	1144	44	18	2793	120.7	1152	45	20
Massif curve, ms clock	3115	135.2	1056	42	16	1557	66.8	1190	49	24
Massif pages-as-heap	3545	153.9	668	39	14	3287	142.3	691	40	15
Massif pages-as-heap, ms clock	5524	241.1	465	34	10	4241	184.8	697	37	12
gperftools peak	9387	410.2	1644	32	7	9993	438.3	1644	31	7
heaptrack curve	10614	464.9	1285	30	6	11443	500.9	1439	29	6

Table 11: Every oracle at offered load 0.6, as Table 3.

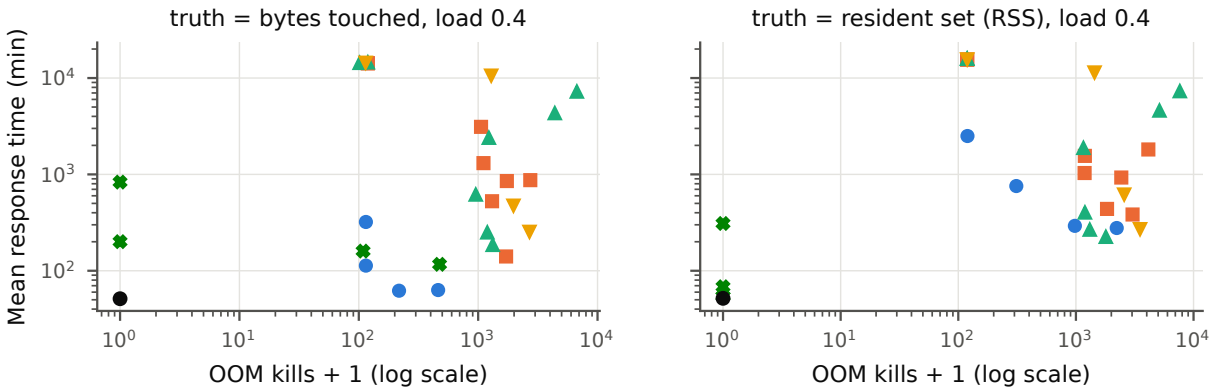
	Truth = bytes touched					Truth = RSS				
Oracle	Resp. (min)	BSLD	Kills	Res. %	Used %	Resp. (min)	BSLD	Kills	Res. %	Used %
Exact curve	56	1.2	0	66	66	56	1.2	0	66	66
Exact, 20 windows	60	1.4	0	72	64	60	1.4	0	71	65
MemPrint windowed 5%	140	4.6	460	72	55	1460	62.6	310	57	32
Exact peak	149	5.1	0	75	57	124	4.0	0	75	59
Dynamic provisioning (5 min)	166	5.9	257	72	61	125	4.2	234	73	62
MemPrint pages only	177	6.2	1900	72	53	2148	92.8	1495	51	26
Reactive resizing (5 min)	264	10.3	12	69	53	272	10.6	17	68	53
Dynamic provisioning, look-back	291	11.5	491	67	56	298	11.9	443	65	55
MemPrint windowed 1%	444	18.0	1116	64	42	1964	84.7	620	52	27
MemPrint windowed 5%, peak	473	19.2	379	71	48	569	23.3	492	70	46
RSS peak (earlier run)	603	24.9	474	67	43	124	4.0	0	75	59
DHAT peak	676	28.1	1323	67	43	931	39.3	1310	64	40
KS+ (segments from other inputs)	941	40.1	8190	59	37	476	19.3	8529	68	49
ML peak (quantile GBM)	1615	69.4	115	62	30	1590	68.3	115	61	31
User request	1833	79.0	0	58	28	1788	77.1	0	58	29
Massif curve	2025	87.2	1106	53	26	1723	74.1	1184	54	29
Pond (untouched in pool)	2131	92.0	0	54	26	2123	91.5	0	53	26
DHAT curve (3 points)	3122	135.5	1228	51	23	2653	115.1	1157	49	23
DHAT hold-then-decay	3775	163.9	1144	46	20	3499	151.7	1152	46	21
Massif curve, ms clock	3857	167.8	1056	44	17	2291	99.2	1190	51	26
Massif pages-as-heap	4246	184.8	668	40	15	4082	177.3	692	39	15
Massif pages-as-heap, ms clock	6263	273.7	465	35	10	5285	230.4	697	37	12
gperftools peak	10144	443.6	1644	32	7	10758	472.2	1644	32	7
heaptrack curve	11432	500.9	1285	31	6	12153	531.8	1439	30	6

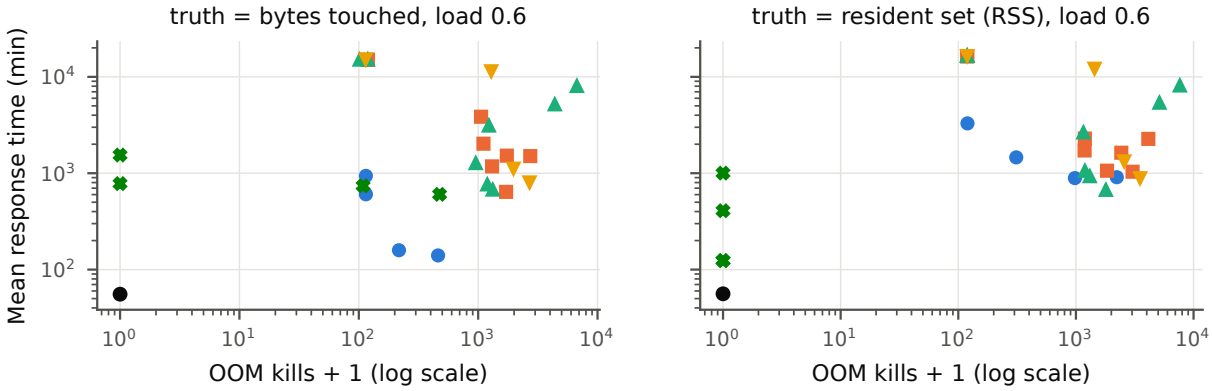
Table 12: Every oracle at offered load 1.0, as Table 3.

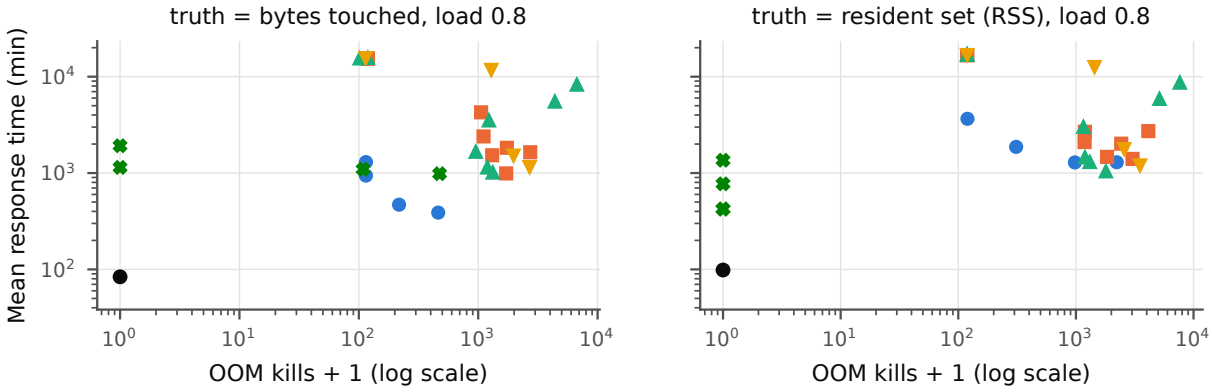
	Truth = bytes touched					Truth = RSS				
Oracle	Resp. (min)	BSLD	Kills	Res. %	Used %	Resp. (min)	BSLD	Kills	Res. %	Used %
Exact curve	258	10.1	0	81	81	284	11.2	0	80	80
Exact, 20 windows	353	14.3	0	83	74	366	14.8	0	81	74
Dynamic provisioning (5 min)	616	25.8	284	78	66	676	28.5	318	76	64
MemPrint windowed 5%	617	25.8	460	78	59	2088	90.1	309	58	33
Exact peak	665	27.9	0	78	59	648	27.1	0	78	61
MemPrint pages only	686	28.6	1902	75	54	2776	120.5	1494	55	28
Dynamic provisioning, look-back	703	29.8	426	72	61	774	32.8	453	71	60
Reactive resizing (5 min)	781	33.1	13	72	56	837	35.6	14	71	55
MemPrint windowed 1%	994	42.5	1117	68	44	2559	111.1	620	55	29
MemPrint windowed 5%, peak	1030	43.7	379	74	50	1188	50.6	492	71	47
RSS peak (earlier run)	1204	51.5	474	69	44	648	27.1	0	78	61
DHAT peak	1271	54.3	1323	69	44	1539	66.1	1310	67	42
KS+ (segments from other inputs)	1398	60.5	8189	61	39	1079	46.1	8531	68	49
ML peak (quantile GBM)	2232	96.5	115	61	30	2224	96.2	115	61	31
User request	2436	105.6	0	59	29	2422	104.9	0	59	30
Massif curve	2611	113.2	1107	55	27	2378	102.9	1184	56	30
Pond (untouched in pool)	2736	118.6	0	55	27	2728	118.2	0	54	27
DHAT curve (3 points)	3732	162.5	1228	54	25	3256	141.7	1157	53	26
DHAT hold-then-decay	4376	190.4	1144	50	21	4165	181.2	1152	49	22
Massif curve, ms clock	4494	195.9	1056	46	18	2933	127.3	1190	52	26
Massif pages-as-heap	4896	213.3	668	42	16	4718	205.2	692	42	17
Massif pages-as-heap, ms clock	7000	305.5	466	35	10	5908	258.0	697	37	13
gperftools peak	10698	468.0	1644	33	8	11382	499.3	1644	33	7
heaptrack curve	11956	523.9	1285	31	6	12829	562.0	1439	32	6

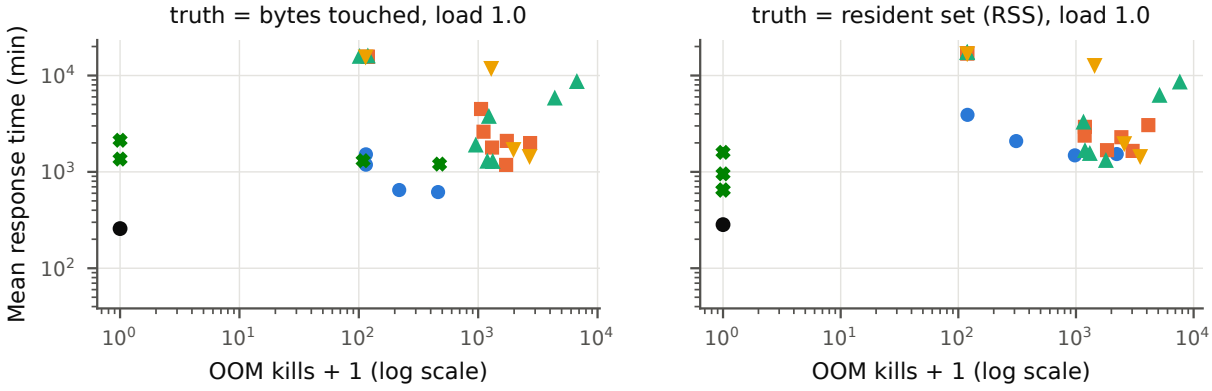
Table 13: Every oracle at offered load 1.2, as Table 3.

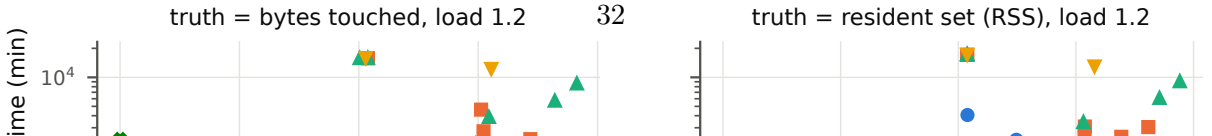
	Truth = bytes touched					Truth = RSS				
Oracle	Resp. (min)	BSLD	Kills	Res. %	Used %	Resp. (min)	BSLD	Kills	Res. %	Used %
Exact curve	386	15.7	0	81	81	414	17.0	0	82	82
Exact, 20 windows	497	20.6	0	82	73	511	21.2	0	82	75
MemPrint windowed 5%	728	30.6	461	79	60	2251	97.4	310	60	34
Dynamic provisioning (5 min)	790	33.5	298	79	67	793	33.6	289	78	67
Dynamic provisioning, look-back	814	34.6	488	77	64	949	40.6	457	73	62
Exact peak	822	34.7	0	78	58	796	33.6	0	77	61
MemPrint pages only	844	35.6	1901	74	53	2938	127.6	1495	57	29
Reactive resizing (5 min)	947	40.4	16	73	56	1006	43.0	17	71	56
MemPrint windowed 1%	1159	49.6	1116	66	43	2723	118.1	620	56	29
MemPrint windowed 5%, peak	1201	51.2	379	74	50	1324	56.6	492	74	49
RSS peak (earlier run)	1333	57.1	474	72	46	796	33.6	0	77	61
DHAT peak	1377	59.1	1323	75	48	1658	71.3	1310	69	44
KS+ (segments from other inputs)	1707	74.2	8201	63	40	1270	54.6	8541	67	47
ML peak (quantile GBM)	2350	101.8	115	63	31	2354	101.9	115	63	32
User request	2581	112.0	0	59	29	2568	111.3	0	61	31
Massif curve	2755	119.5	1107	58	29	2531	109.6	1184	58	32
Pond (untouched in pool)	2909	126.2	0	57	27	2880	124.9	0	57	29
DHAT curve (3 points)	3882	169.2	1228	56	26	3442	149.8	1157	56	27
DHAT hold-then-decay	4532	197.2	1144	53	22	4347	189.0	1152	52	24
Massif curve, ms clock	4625	201.7	1056	48	19	3089	134.2	1190	56	28
Massif pages-as-heap	4996	217.6	668	41	15	4846	210.8	692	42	17
Massif pages-as-heap, ms clock	7190	314.0	466	36	11	6089	265.7	697	39	13
gperftools peak	10858	474.7	1644	33	7	11555	507.0	1644	33	7
heaptrack curve	12219	535.6	1285	31	6	12973	568.1	1439	31	6

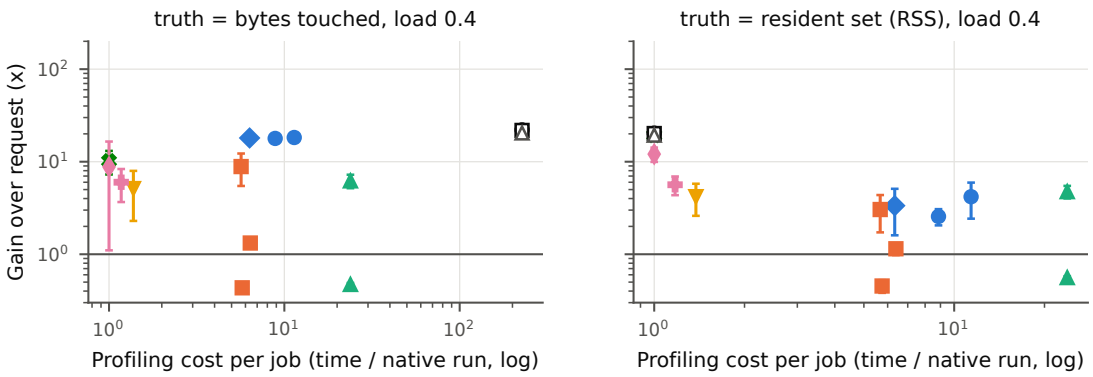

truth = bytes touched, load 0.4
truth = resident set (RSS), load 0.4
Mean response time (min)
OOM kills + 1 (log scale)

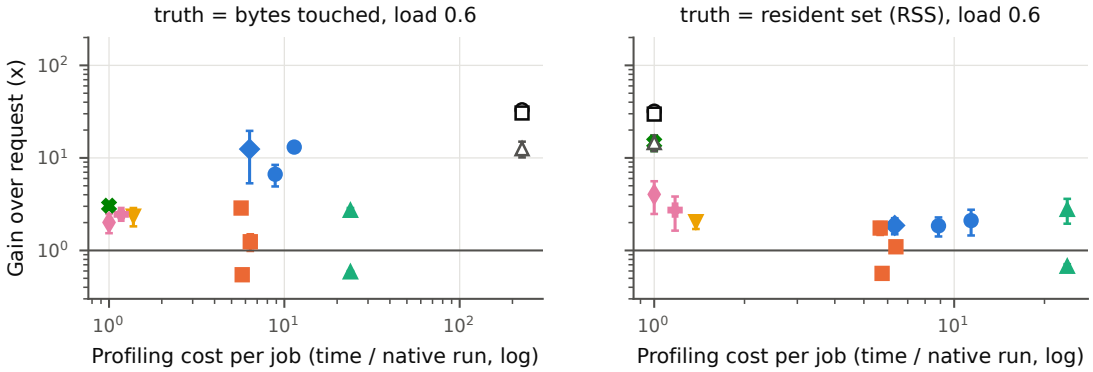

truth = bytes touched, load 0.6
truth = resident set (RSS), load 0.6
Mean response time (min)
OOM kills + 1 (log scale)

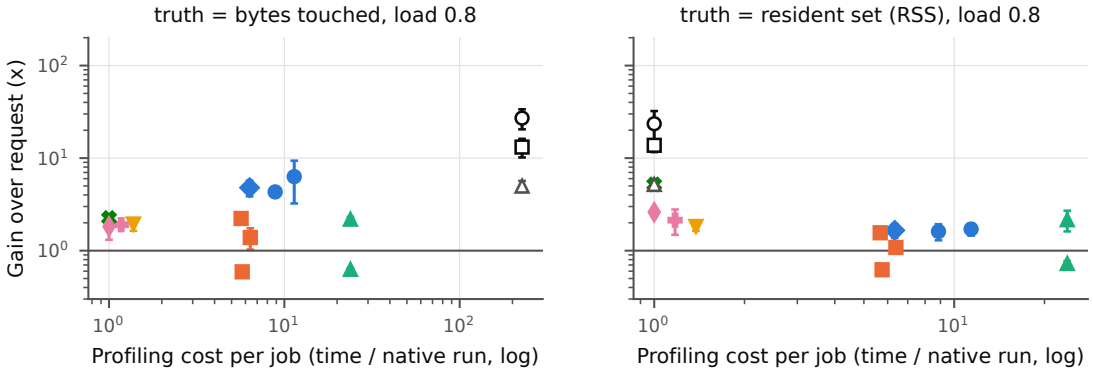

truth = bytes touched, load 0.8
truth = resident set (RSS), load 0.8
Mean response time (min)
OOM kills + 1 (log scale)

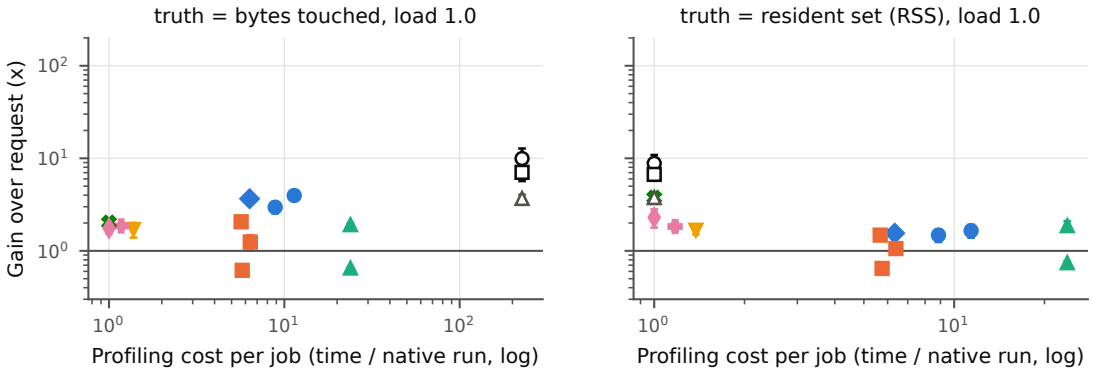

truth = bytes touched, load 1.0
truth = resident set (RSS), load 1.0
Mean response time (min)
OOM kills + 1 (log scale)

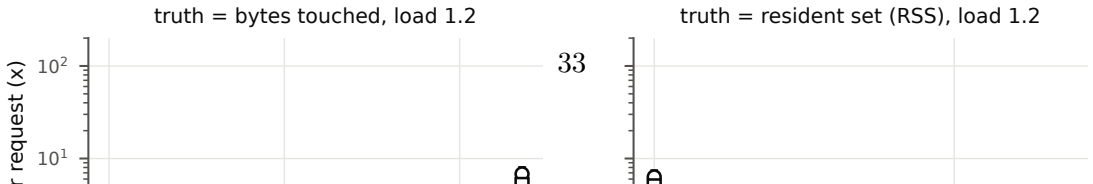

truth = bytes touched, load 1.2
truth = resident set (RSS), load 1.2


truth = bytes touched, load 0.4
truth = resident set (RSS), load 0.4
Gain over request (x)
Profiling cost per job (time / native run, log)


truth = bytes touched, load 0.6
truth = resident set (RSS), load 0.6
Gain over request (x)
Profiling cost per job (time / native run, log)


truth = bytes touched, load 0.8
truth = resident set (RSS), load 0.8
Gain over request (x)
Profiling cost per job (time / native run, log)


truth = bytes touched, load 1.0
truth = resident set (RSS), load 1.0
Gain over request (x)
Profiling cost per job (time / native run, log)


truth = bytes touched, load 1.2
truth = resident set (RSS), load 1.2

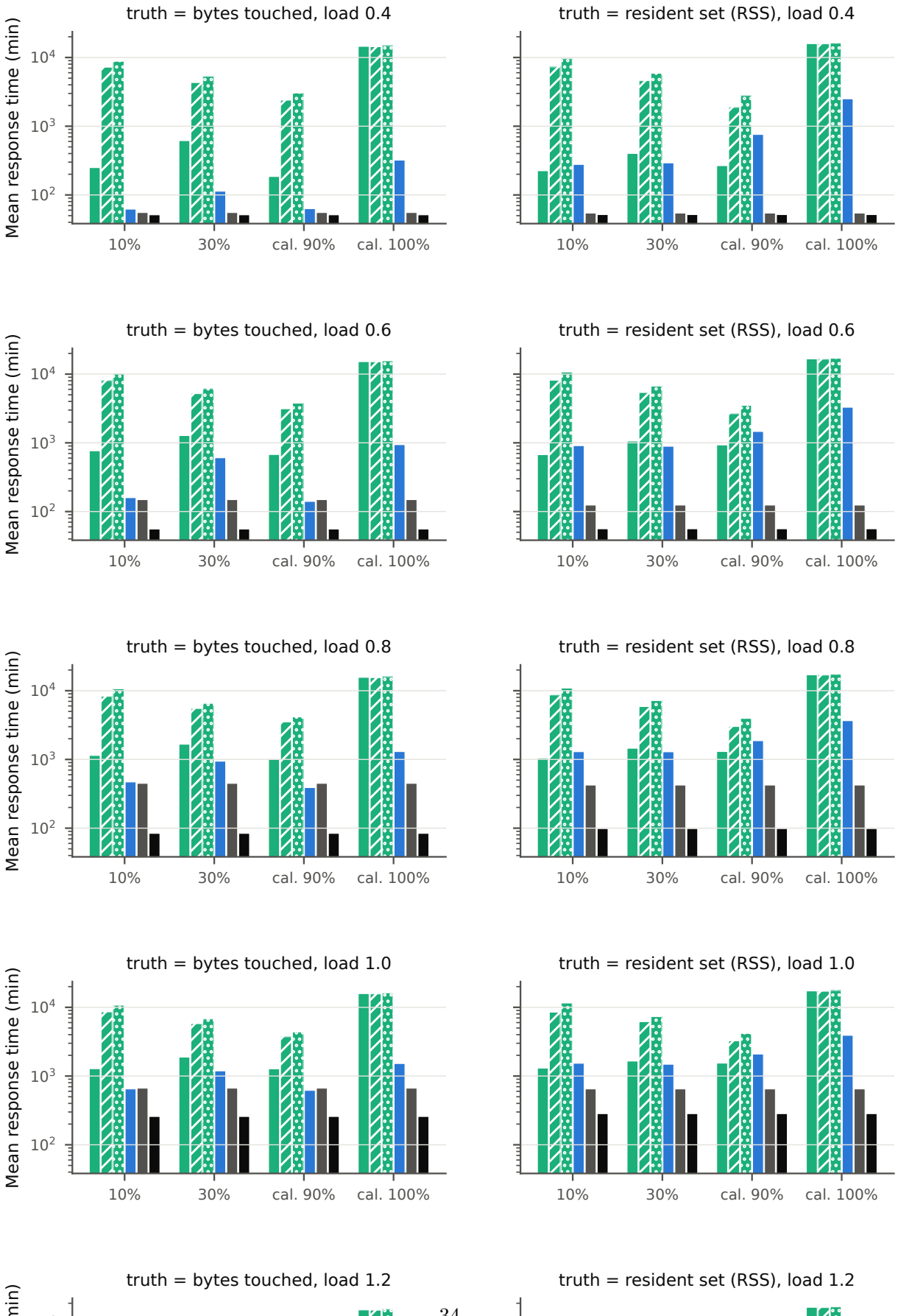

truth = bytes touched, load 0.4
truth = resident set (RSS), load 0.4
truth = bytes touched, load 0.6
truth = resident set (RSS), load 0.6
truth = bytes touched, load 0.8
truth = resident set (RSS), load 0.8
truth = bytes touched, load 1.0
truth = resident set (RSS), load 1.0
truth = bytes touched, load 1.2
truth = resident set (RSS), load 1.2
Mean response time (min)
10^4
10^3
10^2
10%
30%
cal. 90%
cal. 100%

Table 14: The LLM stream at offered load 0.4, as Table 5.

	Truth = bytes touched					Truth = RSS				
Oracle	Resp. (min)	BSLD	Kills	Res. %	Used %	Resp. (min)	BSLD	Kills	Res. %	Used %
Exact curve	53	1.1	0	45	45	53	1.1	0	45	45
Exact, 20 windows	53	1.1	0	47	45	53	1.1	0	47	45
Exact peak	53	1.1	0	50	45	54	1.1	0	52	45
RSS peak (earlier run)	58	1.3	0	54	45	54	1.1	0	52	45
Dynamic provisioning (5 min)	59	1.3	27	50	45	60	1.3	55	50	45
KS+ (segments from other inputs)	64	1.5	1549	52	40	60	1.4	805	52	42
Reactive resizing (5 min)	71	1.8	9	54	45	67	1.7	6	54	45
Dynamic provisioning, look-back	72	1.9	96	50	45	77	2.1	139	50	45
MemPrint windowed 5%, peak	74	1.8	602	59	45	85	2.2	602	62	43
ML peak (quantile GBM)	560	23.0	115	59	35	709	29.5	115	59	33
User request	811	34.1	0	57	33	932	39.4	0	57	32
Pond (untouched in pool)	1101	46.7	0	53	31	1236	52.5	0	53	29
Massif pages-as-heap	1339	57.2	0	48	27	966	40.9	0	50	30
Massif pages-as-heap, ms clock	1660	71.0	0	46	25	1257	53.5	0	49	28
MemPrint windowed 1%	1754	75.6	1048	44	23	1834	78.8	1046	45	22
MemPrint windowed 5%	2787	121.2	1030	40	18	2564	111.7	1032	42	19
MemPrint pages only	5349	234.4	610	29	9	5774	253.4	1220	31	10
heaptrack curve	7412	322.6	1195	30	9	7237	315.0	1195	30	9
Massif curve, ms clock	7453	324.3	1195	30	8	7431	323.9	1195	30	8
Massif curve	7570	329.4	1195	30	8	6956	303.5	1195	31	9
DHAT curve (3 points)	7631	333.2	1195	29	8	7598	331.5	1195	30	8
DHAT peak	7924	345.2	1195	29	8	7717	336.1	1195	30	8
DHAT hold-then-decay	7982	348.3	1195	29	8	7951	347.3	1195	30	8
gperftools peak	8035	350.6	1195	29	8	7867	343.3	1195	29	8

Table 15: The LLM stream at offered load 0.6, as Table 5.

	Truth = bytes touched					Truth = RSS				
Oracle	Resp. (min)	BSLD	Kills	Res. %	Used %	Resp. (min)	BSLD	Kills	Res. %	Used %
Exact curve	64	1.6	0	66	66	64	1.6	0	66	66
Exact, 20 windows	65	1.6	0	68	66	65	1.6	0	68	66
Exact peak	74	1.9	0	71	65	87	2.5	0	73	63
Dynamic provisioning (5 min)	105	3.3	107	69	62	129	4.3	148	68	60
RSS peak (earlier run)	220	8.3	0	67	57	87	2.5	0	73	63
Dynamic provisioning, look-back	252	9.7	199	63	57	362	14.7	316	61	54
KS+ (segments from other inputs)	327	13.1	1537	63	48	190	7.0	800	64	52
Reactive resizing (5 min)	345	13.8	9	64	53	349	14.0	11	64	53
MemPrint windowed 5%, peak	377	14.8	602	67	51	489	19.7	602	69	48
ML peak (quantile GBM)	1404	60.2	115	60	36	1496	64.2	115	60	33
User request	1630	70.1	0	57	33	1715	73.8	0	58	32
Pond (untouched in pool)	1950	83.9	0	53	31	2024	87.2	0	53	29
Massif pages-as-heap	2219	95.9	0	47	26	1782	76.8	0	50	29
Massif pages-as-heap, ms clock	2534	109.5	0	46	25	2085	89.9	0	48	26
MemPrint windowed 1%	2691	116.9	1049	45	23	2616	113.3	1045	47	23
MemPrint windowed 5%	3706	161.4	1035	42	19	3421	149.5	1035	43	19
MemPrint pages only	6238	273.5	610	29	9	6838	300.0	1220	29	9
heaptrack curve	8312	362.1	1195	31	9	8063	351.7	1195	31	9
Massif curve	8426	367.2	1195	30	9	8135	355.1	1195	31	9
Massif curve, ms clock	8509	370.9	1195	31	9	8257	360.1	1195	31	9
DHAT curve (3 points)	8687	379.3	1195	30	8	8345	364.6	1195	31	9
DHAT peak	8835	385.1	1195	30	8	8312	362.1	1195	31	8
gperftools peak	8835	385.8	1195	30	8	8684	379.2	1195	30	8
DHAT hold-then-decay	9023	394.0	1195	30	8	8832	385.7	1195	30	8

Table 16: The LLM stream at offered load 1.0, as Table 5.

	Truth = bytes touched					Truth = RSS				
Oracle	Resp. (min)	BSLD	Kills	Res. %	Used %	Resp. (min)	BSLD	Kills	Res. %	Used %
Exact curve	414	16.9	0	77	77	385	15.6	0	77	77
Exact, 20 windows	447	18.4	0	77	74	439	17.9	0	76	73
Exact peak	547	22.6	0	77	70	593	24.7	0	77	66
Dynamic provisioning (5 min)	674	28.5	152	73	66	690	29.2	226	73	65
Dynamic provisioning, look-back	758	32.1	254	71	64	714	30.3	340	73	66
RSS peak (earlier run)	862	36.5	0	70	59	593	24.7	0	77	66
KS+ (segments from other inputs)	979	42.2	1538	65	50	733	31.1	770	70	56
Reactive resizing (5 min)	995	42.5	11	67	55	952	40.6	12	67	55
MemPrint windowed 5%, peak	1098	46.5	602	68	51	1157	49.1	602	71	48
ML peak (quantile GBM)	2114	91.3	115	60	35	2197	95.0	115	60	33
User request	2351	101.8	0	57	33	2396	103.7	0	57	31
Pond (untouched in pool)	2673	115.7	0	53	31	2705	117.2	0	53	29
Massif pages-as-heap	2918	126.6	0	48	27	2455	106.3	0	51	29
Massif pages-as-heap, ms clock	3228	140.0	0	46	25	2760	119.6	0	49	27
MemPrint windowed 1%	3316	144.3	1049	48	24	3222	139.7	1045	50	25
MemPrint windowed 5%	4459	194.6	1034	43	20	4065	177.9	1032	45	20
MemPrint pages only	6945	304.7	610	29	9	7491	328.7	1220	30	9
Massif curve	9086	396.2	1195	32	9	8765	382.8	1195	32	10
heaptrack curve	9205	401.1	1195	32	9	8754	381.6	1195	33	10
Massif curve, ms clock	9225	402.3	1195	32	10	8964	391.1	1195	33	10
DHAT curve (3 points)	9343	408.2	1195	32	9	9034	395.0	1195	32	9
DHAT peak	9606	418.9	1195	32	9	9292	405.1	1195	33	9
gperftools peak	9634	420.8	1195	31	9	9371	409.1	1195	32	9
DHAT hold-then-decay	9842	430.0	1195	31	9	9509	415.5	1195	32	9

Table 17: The LLM stream at offered load 1.2, as Table 5.

	Truth = bytes touched					Truth = RSS				
Oracle	Resp. (min)	BSLD	Kills	Res. %	Used %	Resp. (min)	BSLD	Kills	Res. %	Used %
Exact curve	603	25.3	0	76	76	550	22.9	0	77	77
Exact, 20 windows	629	26.4	0	77	74	577	24.0	0	78	75
Exact peak	721	30.2	0	77	69	769	32.4	0	77	66
Dynamic provisioning (5 min)	864	36.8	159	75	68	891	38.0	230	73	65
Dynamic provisioning, look-back	921	39.4	287	74	66	883	37.7	386	76	68
RSS peak (earlier run)	1042	44.4	0	70	58	769	32.4	0	77	66
KS+ (segments from other inputs)	1182	51.1	1531	65	50	969	41.4	785	69	55
Reactive resizing (5 min)	1192	51.1	9	67	56	1120	48.0	12	69	57
MemPrint windowed 5%, peak	1226	52.2	602	71	53	1350	57.6	602	72	48
ML peak (quantile GBM)	2270	98.2	115	60	35	2342	101.4	115	61	33
User request	2546	110.3	0	58	34	2583	111.9	0	59	32
Pond (untouched in pool)	2833	122.8	0	56	32	2848	123.5	0	55	30
Massif pages-as-heap	3086	134.0	0	49	28	2626	113.8	0	53	31
MemPrint windowed 1%	3406	148.6	1045	49	25	3561	154.7	1047	51	25
Massif pages-as-heap, ms clock	3427	148.8	0	47	25	2942	127.5	0	50	28
MemPrint windowed 5%	4533	198.5	1032	46	21	4247	185.6	1033	46	20
MemPrint pages only	7155	313.9	610	29	9	7613	334.2	1220	31	10
Massif curve	9216	402.1	1195	33	10	8974	392.0	1195	33	10
Massif curve, ms clock	9382	409.3	1195	32	9	9082	396.5	1195	33	10
heaptrack curve	9387	409.1	1195	32	10	8757	381.9	1195	33	10
DHAT curve (3 points)	9613	420.0	1195	33	10	9333	407.8	1195	33	10
gperftools peak	9761	426.5	1195	32	10	9544	416.8	1195	33	9
DHAT hold-then-decay	9841	430.2	1195	32	9	9763	426.6	1195	32	9
DHAT peak	9842	429.0	1195	31	9	9495	414.0	1195	33	10

Table 18: MemPrint's oracles before and after tracking file mappings, at offered load 0.4, as Table 8.

	Main stream (25 programs)				LLM stream			
	Bytes touched		RSS		Bytes touched		RSS	
Oracle	before	after	before	after	before	after	before	after
MemPrint windowed 5%	63	59	757	807	2787	57	2564	104
MemPrint windowed 1%	71	60	1199	753	1754	57	1834	113
MemPrint pages only	65	63	1404	784	5349	57	5774	102
MemPrint windowed 5%, peak	79	75	114	105	74	72	85	135

Table 19: MemPrint's oracles before and after tracking file mappings, at offered load 0.6, as Table 8.

	Main stream (25 programs)				LLM stream			
	Bytes touched		RSS		Bytes touched		RSS	
Oracle	before	after	before	after	before	after	before	after
MemPrint windowed 5%	140	102	1460	1485	3706	123	3421	502
MemPrint windowed 1%	444	121	1964	1456	2691	120	2616	541
MemPrint pages only	177	161	2148	1469	6238	117	6838	548
MemPrint windowed 5%, peak	473	448	569	598	377	278	489	626

Table 20: MemPrint's oracles before and after tracking file mappings, at offered load 1.0, as Table 8.

	Main stream (25 programs)				LLM stream			
	Bytes touched		RSS		Bytes touched		RSS	
Oracle	before	after	before	after	before	after	before	after
MemPrint windowed 5%	617	573	2088	2127	4459	613	4065	1201
MemPrint windowed 1%	994	634	2559	2088	3316	603	3222	1155
MemPrint pages only	686	621	2776	2104	6945	620	7491	1183
MemPrint windowed 5%, peak	1030	1010	1188	1204	1098	933	1157	1201

Table 21: MemPrint's oracles before and after tracking file mappings, at offered load 1.2, as Table 8.

	Main stream (25 programs)				LLM stream			
	Bytes touched		RSS		Bytes touched		RSS	
Oracle	before	after	before	after	before	after	before	after
MemPrint windowed 5%	728	708	2251	2284	4533	795	4247	1346
MemPrint windowed 1%	1159	738	2723	2259	3406	804	3561	1329
MemPrint pages only	844	744	2938	2262	7155	760	7613	1308
MemPrint windowed 5%, peak	1201	1158	1324	1332	1226	1095	1350	1426

Table 22: Mean response time under three truths at offered load 0.4, as Table 9.

Oracle	Truth = bytes touched	Truth = RSS	Truth = RSS without code
Exact curve	51	52	52
Exact peak	55	54	55
RSS peak (earlier run)	117	54	55
MemPrint windowed 5%	59	807	849
MemPrint 5%, native clock	560	68	66
MemPrint pages, native clock	534	80	72
MemPrint 5%, native clock, RSS level	958	66	67
MemPrint pages, native clock, RSS level	983	68	66
DHAT peak	185	266	216
Massif curve	1,309	1,034	1,195
heaptrack curve	10,614	11,443	11,216
gperftools peak	9,387	9,993	9,893
User request	1,128	1,051	1,084

Table 23: Mean response time under three truths at offered load 0.6, as Table 9.

Oracle	Truth = bytes touched	Truth = RSS	Truth = RSS without code
Exact curve	56	56	56
Exact peak	149	124	134
RSS peak (earlier run)	603	124	134
MemPrint windowed 5%	102	1,485	1,510
MemPrint 5%, native clock	1,180	331	221
MemPrint pages, native clock	1,169	404	318
MemPrint 5%, native clock, RSS level	1,672	250	161
MemPrint pages, native clock, RSS level	1,664	243	167
DHAT peak	676	931	829
Massif curve	2,025	1,723	1,885
heaptrack curve	11,432	12,153	12,007
gperftools peak	10,144	10,758	10,706
User request	1,833	1,788	1,815

Table 24: Mean response time under three truths at offered load 1.0, as Table 9.

Oracle	Truth = bytes touched	Truth = RSS	Truth = RSS without code
Exact curve	258	284	257
Exact peak	665	648	649
RSS peak (earlier run)	1,204	648	649
MemPrint windowed 5%	573	2,127	2,131
MemPrint 5%, native clock	1,811	908	786
MemPrint pages, native clock	1,777	1,016	895
MemPrint 5%, native clock, RSS level	2,275	818	662
MemPrint pages, native clock, RSS level	2,276	819	729
DHAT peak	1,271	1,539	1,445
Massif curve	2,611	2,378	2,516
heaptrack curve	11,956	12,829	12,607
gperftools peak	10,698	11,382	11,285
User request	2,436	2,422	2,439

Table 25: Mean response time under three truths at offered load 1.2, as Table 9.

Oracle	Truth = bytes touched	Truth = RSS	Truth = RSS without code
Exact curve	386	414	407
Exact peak	822	796	811
RSS peak (earlier run)	1,333	796	811
MemPrint windowed 5%	708	2,284	2,298
MemPrint 5%, native clock	1,946	1,042	939
MemPrint pages, native clock	1,943	1,195	1,028
MemPrint 5%, native clock, RSS level	2,402	992	844
MemPrint pages, native clock, RSS level	2,418	987	816
DHAT peak	1,377	1,658	1,576
Massif curve	2,755	2,531	2,693
heaptrack curve	12,219	12,973	12,790
gperftools peak	10,858	11,555	11,445
User request	2,581	2,568	2,585

Table 26: Total memory (TB: 32 nodes of local DRAM plus the 1 TB pool) each oracle needs to keep the mean bounded slowdown within 1.5 times the exact curve's at 128 GB per node, at every offered load (the target is set at each load). The sweep runs from 48 to 512 GB per node; – means not met at 512 GB.

	Truth = bytes touched					Truth = resident set (RSS)				
Load	0.4	0.6	0.8	1	1.2	0.4	0.6	0.8	1	1.2
Exact curve	4.0	5.0	5.0	5.0	5.0	4.0	5.0	5.0	5.0	5.0
Exact, 20 windows	4.0	5.0	6.0	5.0	5.0	4.0	5.0	6.0	5.0	5.0
Exact peak	5.0	6.0	6.0	6.0	6.0	5.0	6.0	6.0	6.0	6.0
MemPrint pages only	5.0	6.0	7.0	6.0	6.0	9.0	11.0	11.0	9.0	9.0
Reactive resizing (5 min)	6.0	6.0	7.0	7.0	6.0	5.0	6.0	7.0	7.0	6.0
Dynamic provisioning (5 min)	5.0	6.0	7.0	6.0	6.0	5.0	6.0	6.0	6.0	6.0
MemPrint windowed 5%	5.0	6.0	7.0	6.0	6.0	7.0	9.0	9.0	9.0	9.0
Dynamic provisioning, look-back	6.0	7.0	7.0	6.0	7.0	6.0	7.0	9.0	6.0	6.0
RSS peak (earlier run)	6.0	7.0	9.0	7.0	7.0	5.0	6.0	6.0	6.0	6.0
ML peak (quantile GBM)	9.0	9.0	9.0	9.0	9.0	9.0	9.0	9.0	9.0	9.0
KS+ (segments from other inputs)	6.0	9.0	9.0	9.0	9.0	6.0	9.0	9.0	9.0	9.0
DHAT peak	6.0	7.0	9.0	7.0	7.0	7.0	9.0	9.0	9.0	9.0
User request	9.0	9.0	11.0	9.0	9.0	9.0	9.0	9.0	9.0	9.0
Massif curve	9.0	11.0	11.0	11.0	9.0	9.0	9.0	9.0	9.0	9.0
DHAT curve (3 points)	11.0	13.0	13.0	13.0	13.0	11.0	11.0	11.0	11.0	11.0
Massif curve, ms clock	11.0	13.0	17.0	13.0	13.0	9.0	11.0	11.0	11.0	11.0
Pond (untouched in pool)	11.0	13.0	17.0	13.0	13.0	11.0	13.0	17.0	13.0	13.0
gperftools peak	–	–	–	–	–	–	–	–	–	–
heaptrack curve	–	–	–	–	–	–	–	–	–	–

Table 27: The same for the oracles of the section on the time axis and the level, under three truths.

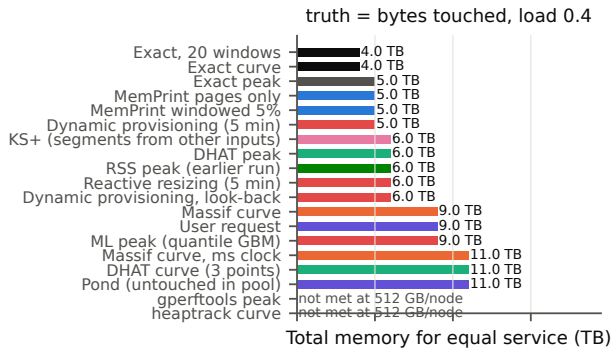
	Truth = bytes touched					Truth = resident set (RSS)					Truth = RSS without code				
Load	0.4	0.6	0.8	1	1.2	0.4	0.6	0.8	1	1.2	0.4	0.6	0.8	1	1.2
Exact curve	4.0	5.0	5.0	5.0	5.0	4.0	5.0	5.0	5.0	5.0	4.0	5.0	5.0	5.0	5.0
Exact peak	5.0	6.0	6.0	6.0	6.0	5.0	6.0	6.0	6.0	6.0	5.0	6.0	6.0	6.0	6.0
MemPrint windowed 5%	5.0	6.0	6.0	6.0	6.0	7.0	9.0	9.0	9.0	9.0	7.0	9.0	9.0	9.0	9.0
DHAT peak	6.0	7.0	9.0	7.0	7.0	7.0	9.0	9.0	9.0	9.0	7.0	9.0	9.0	9.0	7.0
MemPrint 5%, native clock	7.0	9.0	9.0	9.0	9.0	6.0	6.0	7.0	7.0	6.0	5.0	6.0	7.0	7.0	6.0
RSS peak (earlier run)	6.0	7.0	9.0	7.0	7.0	5.0	6.0	6.0	6.0	6.0	5.0	6.0	6.0	6.0	6.0
MemPrint 5%, native clock, RSS level	9.0	9.0	11.0	9.0	9.0	5.0	6.0	7.0	6.0	6.0	5.0	6.0	7.0	6.0	6.0

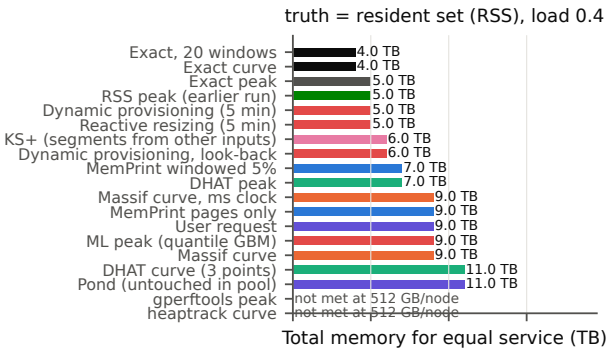
Table 28: The same for MemPrint's curves with file mappings tracked, held to the targets of Table 26.

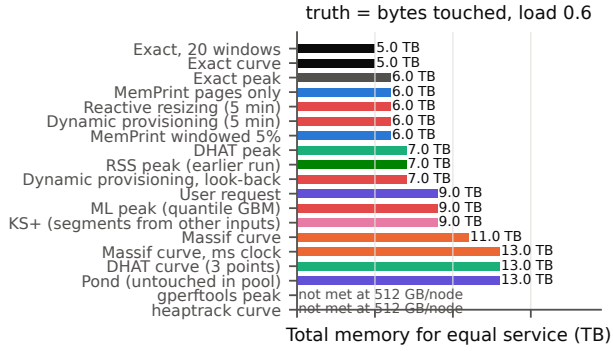
	Truth = bytes touched					Truth = resident set (RSS)				
Load	0.4	0.6	0.8	1	1.2	0.4	0.6	0.8	1	1.2
MemPrint windowed 5%	5.0	6.0	6.0	6.0	6.0	7.0	9.0	9.0	9.0	9.0
MemPrint pages only	5.0	6.0	7.0	6.0	6.0	7.0	9.0	9.0	9.0	9.0

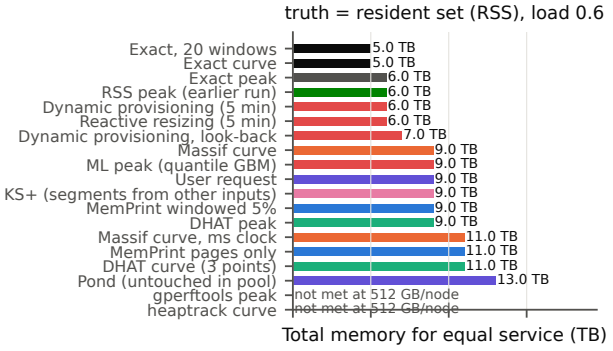
Table 29: Mean response time (minutes) at offered load 0.8 as the remote-memory slowdown β varies (default 0.5), calibrated 90% margins, mean of three seeds.

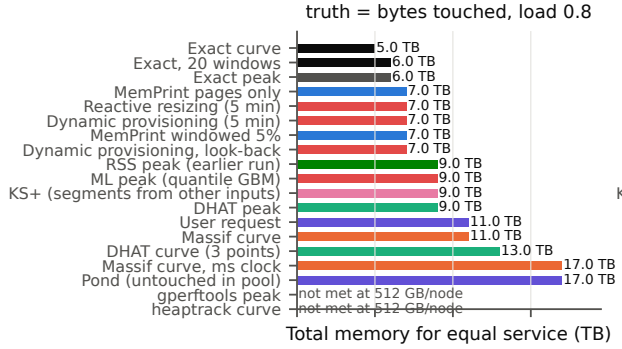
	Truth = bytes touched					Truth = RSS				
β	0.1	0.25	0.5	1	2	0.1	0.25	0.5	1	2
Exact curve	58	63	84	190	390	58	67	99	201	470
Exact peak	274	340	448	693	1,162	233	302	423	666	1,194
MemPrint windowed 5%	236	302	389	598	825	1,491	1,642	1,869	2,239	3,063
MemPrint pages only	363	419	469	630	952	2,147	2,284	2,516	3,058	3,951
DHAT peak	775	927	1,011	1,360	1,975	1,001	1,122	1,302	1,634	2,379
User request	1,896	2,012	2,208	2,601	3,365	1,846	1,977	2,183	2,627	3,431
Reactive resizing (5 min)	432	500	580	779	1,273	446	497	612	838	1,343
Pond (untouched in pool)	2,188	2,301	2,519	2,903	3,649	2,160	2,271	2,472	2,925	3,748
Dynamic provisioning (5 min)	286	332	457	568	1,021	274	363	395	594	932
Dynamic provisioning, look-back	481	439	504	686	1,121	460	507	493	706	1,064
KS+ (segments from other inputs)	1,031	963	1,262	1,378	1,828	704	741	844	1,215	1,282

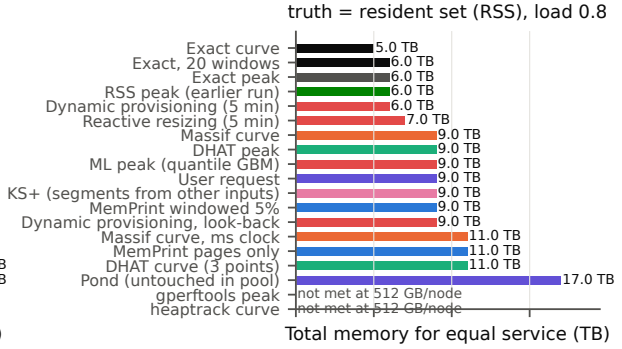

truth = bytes touched, load 0.4
Exact, 20 windows
Exact curve
Exact peak
MemPrint pages only
MemPrint windowed 5%
Dynamic provisioning (5 min)
KS+ (segments from other inputs)
DHAT peak
RSS peak (earlier run)
Reactive resizing (5 min)
Dynamic provisioning, look-back
Massif curve
User request
ML peak (quantile GBM)
Massif curve, ms clock
DHAT curve (3 points)
Pond (untouched in pool)
gperftools peak
heaptrack curve
4.0 TB
4.0 TB
5.0 TB
5.0 TB
5.0 TB
5.0 TB
6.0 TB
6.0 TB
6.0 TB
6.0 TB
6.0 TB
9.0 TB
9.0 TB
9.0 TB
11.0 TB
11.0 TB
11.0 TB
not met at 512 GB/node
not met at 512 GB/node
Total memory for equal service (TB)

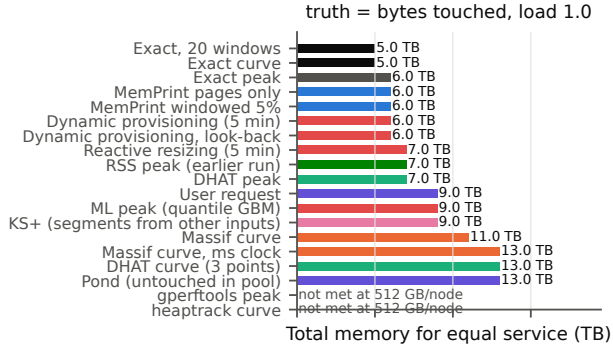

truth = resident set (RSS), load 0.4
Exact, 20 windows
Exact curve
Exact peak
RSS peak (earlier run)
Dynamic provisioning (5 min)
Reactive resizing (5 min)
KS+ (segments from other inputs)
Dynamic provisioning, look-back
MemPrint windowed 5%
DHAT peak
Massif curve, ms clock
MemPrint pages only
User request
ML peak (quantile GBM)
Massif curve
DHAT curve (3 points)
Pond (untouched in pool)
gperftools peak
heaptrack curve
4.0 TB
4.0 TB
5.0 TB
5.0 TB
5.0 TB
5.0 TB
6.0 TB
6.0 TB
7.0 TB
7.0 TB
9.0 TB
9.0 TB
9.0 TB
9.0 TB
9.0 TB
11.0 TB
11.0 TB
not met at 512 GB/node
not met at 512 GB/node
Total memory for equal service (TB)

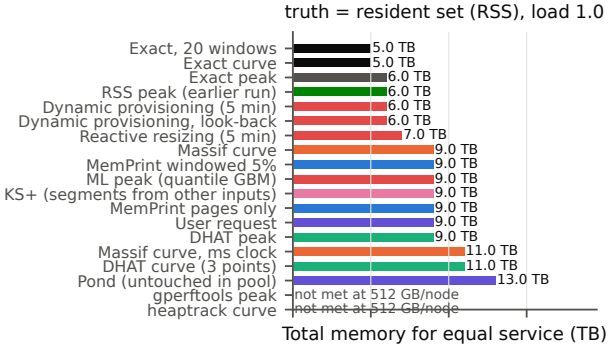

truth = bytes touched, load 0.6
Exact, 20 windows
Exact curve
Exact peak
MemPrint pages only
Reactive resizing (5 min)
Dynamic provisioning (5 min)
MemPrint windowed 5%
DHAT peak
RSS peak (earlier run)
Dynamic provisioning, look-back
User request
ML peak (quantile GBM)
KS+ (segments from other inputs)
Massif curve
Massif curve, ms clock
DHAT curve (3 points)
Pond (untouched in pool)
gperftools peak
heaptrack curve
5.0 TB
5.0 TB
6.0 TB
6.0 TB
6.0 TB
6.0 TB
6.0 TB
7.0 TB
7.0 TB
7.0 TB
9.0 TB
9.0 TB
9.0 TB
11.0 TB
13.0 TB
13.0 TB
13.0 TB
not met at 512 GB/node
not met at 512 GB/node
Total memory for equal service (TB)

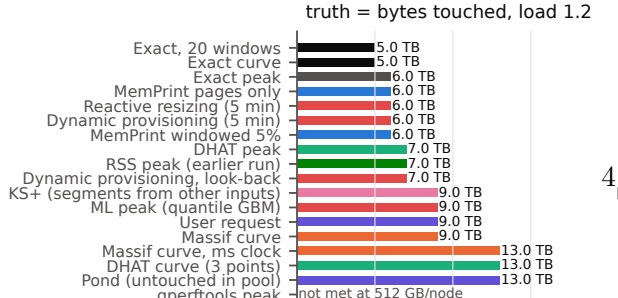

truth = resident set (RSS), load 0.6
Exact, 20 windows
Exact curve
Exact peak
RSS peak (earlier run)
Dynamic provisioning (5 min)
Reactive resizing (5 min)
Dynamic provisioning, look-back
Massif curve
ML peak (quantile GBM)
User request
KS+ (segments from other inputs)
MemPrint windowed 5%
DHAT peak
Massif curve, ms clock
MemPrint pages only
DHAT curve (3 points)
Pond (untouched in pool)
gperftools peak
heaptrack curve
5.0 TB
5.0 TB
6.0 TB
6.0 TB
6.0 TB
6.0 TB
7.0 TB
9.0 TB
9.0 TB
9.0 TB
9.0 TB
9.0 TB
9.0 TB
11.0 TB
11.0 TB
11.0 TB
13.0 TB
not met at 512 GB/node
not met at 512 GB/node
Total memory for equal service (TB)


truth = bytes touched, load 0.8
Exact curve
Exact, 20 windows
Exact peak
MemPrint pages only
Reactive resizing (5 min)
Dynamic provisioning (5 min)
MemPrint windowed 5%
Dynamic provisioning, look-back
RSS peak (earlier run)
ML peak (quantile GBM)
KS+ (segments from other inputs)
DHAT peak
User request
Massif curve
DHAT curve (3 points)
Massif curve, ms clock
Pond (untouched in pool)
gperftools peak
heaptrack curve
5.0 TB
6.0 TB
6.0 TB
7.0 TB
7.0 TB
7.0 TB
7.0 TB
7.0 TB
9.0 TB
9.0 TB
9.0 TB
9.0 TB
11.0 TB
11.0 TB
13.0 TB
17.0 TB
17.0 TB
not met at 512 GB/node
not met at 512 GB/node
Total memory for equal service (TB)


truth = resident set (RSS), load 0.8
Exact curve
Exact, 20 windows
Exact peak
RSS peak (earlier run)
Dynamic provisioning (5 min)
Reactive resizing (5 min)
Massif curve
DHAT peak
ML peak (quantile GBM)
User request
KS+ (segments from other inputs)
MemPrint windowed 5%
Dynamic provisioning, look-back
Massif curve, ms clock
MemPrint pages only
DHAT curve (3 points)
Pond (untouched in pool)
gperftools peak
heaptrack curve
5.0 TB
6.0 TB
6.0 TB
6.0 TB
6.0 TB
7.0 TB
9.0 TB
9.0 TB
9.0 TB
9.0 TB
9.0 TB
9.0 TB
9.0 TB
11.0 TB
11.0 TB
11.0 TB
17.0 TB
not met at 512 GB/node
not met at 512 GB/node
Total memory for equal service (TB)


truth = bytes touched, load 1.0
Exact, 20 windows
Exact curve
Exact peak
MemPrint pages only
MemPrint windowed 5%
Dynamic provisioning (5 min)
Dynamic provisioning, look-back
Reactive resizing (5 min)
RSS peak (earlier run)
DHAT peak
User request
ML peak (quantile GBM)
KS+ (segments from other inputs)
Massif curve
Massif curve, ms clock
DHAT curve (3 points)
Pond (untouched in pool)
gperftools peak
heaptrack curve
5.0 TB
5.0 TB
6.0 TB
6.0 TB
6.0 TB
6.0 TB
6.0 TB
7.0 TB
7.0 TB
7.0 TB
9.0 TB
9.0 TB
9.0 TB
11.0 TB
13.0 TB
13.0 TB
13.0 TB
not met at 512 GB/node
not met at 512 GB/node
Total memory for equal service (TB)


truth = resident set (RSS), load 1.0
Exact, 20 windows
Exact curve
Exact peak
RSS peak (earlier run)
Dynamic provisioning (5 min)
Dynamic provisioning, look-back
Reactive resizing (5 min)
Massif curve
MemPrint windowed 5%
ML peak (quantile GBM)
KS+ (segments from other inputs)
MemPrint pages only
User request
DHAT peak
Massif curve, ms clock
DHAT curve (3 points)
Pond (untouched in pool)
gperftools peak
heaptrack curve
5.0 TB
5.0 TB
6.0 TB
6.0 TB
6.0 TB
6.0 TB
6.0 TB
7.0 TB
9.0 TB
9.0 TB
9.0 TB
9.0 TB
9.0 TB
9.0 TB
9.0 TB
11.0 TB
11.0 TB
13.0 TB
not met at 512 GB/node
not met at 512 GB/node
Total memory for equal service (TB)


truth = bytes touched, load 1.2
Exact, 20 windows
Exact curve
Exact peak
MemPrint pages only
Reactive resizing (5 min)
Dynamic provisioning (5 min)
MemPrint windowed 5%
DHAT peak
RSS peak (earlier run)
Dynamic provisioning, look-back
KS+ (segments from other inputs)
ML peak (quantile GBM)
User request
Massif curve
Massif curve, ms clock
DHAT curve (3 points)
Pond (untouched in pool)
gperftools peak
5.0 TB
5.0 TB
6.0 TB
6.0 TB
6.0 TB
6.0 TB
6.0 TB
7.0 TB
7.0 TB
7.0 TB
9.0 TB
9.0 TB
9.0 TB
9.0 TB
13.0 TB
13.0 TB
13.0 TB

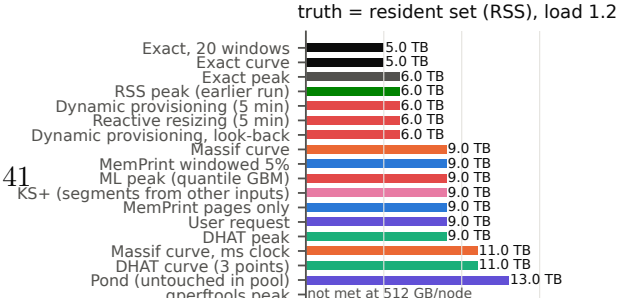

truth = resident set (RSS), load 1.2
Exact, 20 windows
Exact curve
Exact peak
RSS peak (earlier run)
Dynamic provisioning (5 min)
Reactive resizing (5 min)
Dynamic provisioning, look-back
Massif curve
MemPrint windowed 5%
ML peak (quantile GBM)
KS+ (segments from other inputs)
MemPrint pages only
User request
DHAT peak
Massif curve, ms clock
DHAT curve (3 points)
Pond (untouched in pool)
gperftools peak
5.0 TB
5.0 TB
6.0 TB
6.0 TB
6.0 TB
6.0 TB
6.0 TB
9.0 TB
9.0 TB
9.0 TB
9.0 TB
9.0 TB
9.0 TB
9.0 TB
11.0 TB
11.0 TB
13.0 TB

Table 30: Mean response time (minutes) at offered load 0.8 as the pool varies (default 1 TB). Without a pool, the 569 jobs per stream larger than a node are rejected, so that case is left out; a 512 GB pool rejects 11.

	Truth = bytes touched				Truth = RSS			
Pool (GB)	512	1024	2048	4096	512	1024	2048	4096
Exact curve	565	84	51	51	671	99	51	51
Exact peak	1,499	448	78	52	1,455	423	65	52
MemPrint windowed 5%	1,503	389	77	54	3,734	1,869	563	128
MemPrint pages only	1,444	469	108	55	4,580	2,516	871	217
DHAT peak	2,354	1,011	262	65	2,818	1,302	354	78
User request	4,441	2,208	697	261	4,458	2,183	667	222
Reactive resizing (5 min)	1,669	580	94	52	1,708	612	90	52
Pond (untouched in pool)	4,982	2,519	802	258	4,976	2,472	770	221
Dynamic provisioning (5 min)	1,624	457	58	50	1,617	395	57	50
Dynamic provisioning, look-back	1,887	504	56	50	2,134	493	56	50
KS+ (segments from other inputs)	2,005	1,262	437	126	1,669	844	350	113

Table 31: Mean response time (minutes) at offered load 0.8 as the update interval of the three policies that resize a running job varies (default 5 minutes).

	Truth = bytes touched				Truth = RSS			
Update interval (min)	1	5	15	30	1	5	15	30
Reactive resizing	321	580	1,034	1,402	329	612	1,063	1,421
Dynamic provisioning	157	457	812	1,330	182	395	872	1,232
Dynamic provisioning, look-back	182	504	1,175	1,657	191	493	1,029	1,758

with one every 30 minutes under the bytes-touched truth, against 389 for MemPrint's curve, which needs no updates; its look-back variant gives 182 to 1,657 minutes. A site that can resize jobs every minute gets close to the exact curve with dynamic provisioning; one that resizes every 15 minutes or more loses most of its advantage.

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